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Auditory, Vestibular, and Visual Impairments

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SUMMARY PEARLS

Auditory Impairments

- Processing of complex auditory information largely takes place in the auditory pathways and center within the brainstem, thalamus, and cerebral cortex.
- Pure-tone audiometry is typically included in the initial test battery for the hearing assessment of older children and adults. However, it has limitations as a clinical measure of auditory function, particularly in patients with cognitive and/or language impairment.
- Speech audiometry procedures, such as speech perception in noise tests, are useful in evaluating central auditory nervous system function.
- Objective auditory tests, such as otoacoustic emissions and the auditory brainstem response, permit valid assessment of hearing in infants, young children, and patients with cognitive impairment.
- Measurement of the auditory brainstem response is the clinical technique of choice for hearing screening of newborn infants and for diagnosis of hearing loss in young children.
- Outer hair cells within the cochlea have the unique characteristic of motility. Up-and-down movement of the outer hair cells in response to sound stimulation creates additional energy in the cochlea, contributing to improved hearing ability.
- The otoacoustic emissions test, an objective clinical procedure for measuring outer hair cell integrity, can be used in screening for hearing loss in children and for diagnosis of auditory dysfunction.

Vestibular Impairments

- Vestibular system dysfunction can lead to a multitude of problems, including dizziness, vertigo, disequilibrium, nystagmus, oscillopsia, and increased risk for falls, all of which adversely affect participation and clinical outcome of the rehabilitation process.
- Both the peripheral and central vestibular systems contribute to the functions of balance, equilibrium, orientation in space, ocular stability during movement, and maintenance of posture.
- The abnormal migration of otoconia into the semicircular canal system results in benign paroxysmal positional vertigo.
- The Dix-Hallpike maneuver is often used to assess for the effects of displaced otoconia in the posterior semicircular canal.
- Vestibular rehabilitation is an exercise-based therapy that enhances central compensation, leading to improvements in symptoms such as imbalance, actual falls, oscillopsia, dizziness, vertigo, nausea, anxiety, and fear of falling.
- There is strong evidence to support the use of vestibular physical therapy in alleviating vestibular symptoms such as dizziness and vertigo and in improving gaze and postural stability. As in all other sensory impairments, early intervention is recommended.

Visual Impairments

- Noncorrectible visual impairments affect over 7 million people in the United States.
- Visual impairment commonly results from age-related eye disease or acquired brain injury with vision loss ranging from mild visual impairment to total blindness.
- Vision rehabilitation focuses on restitutive, compensatory, and substitution strategies to maximize an individual's functioning with the visual loss.
- Any severity of vision loss may result in negative psychological and social consequences, loss of financial independence, and inability to function independently in daily life. Low vision and binocular vision services are effective in restoring an individual's independence and quality of life.
- Among all age groups, dual sensory impairment is most prevalent in persons 80 years of age and older but is also a growing concern for the younger post-9/11 veteran population.

Auditory Impairments and Hearing Loss

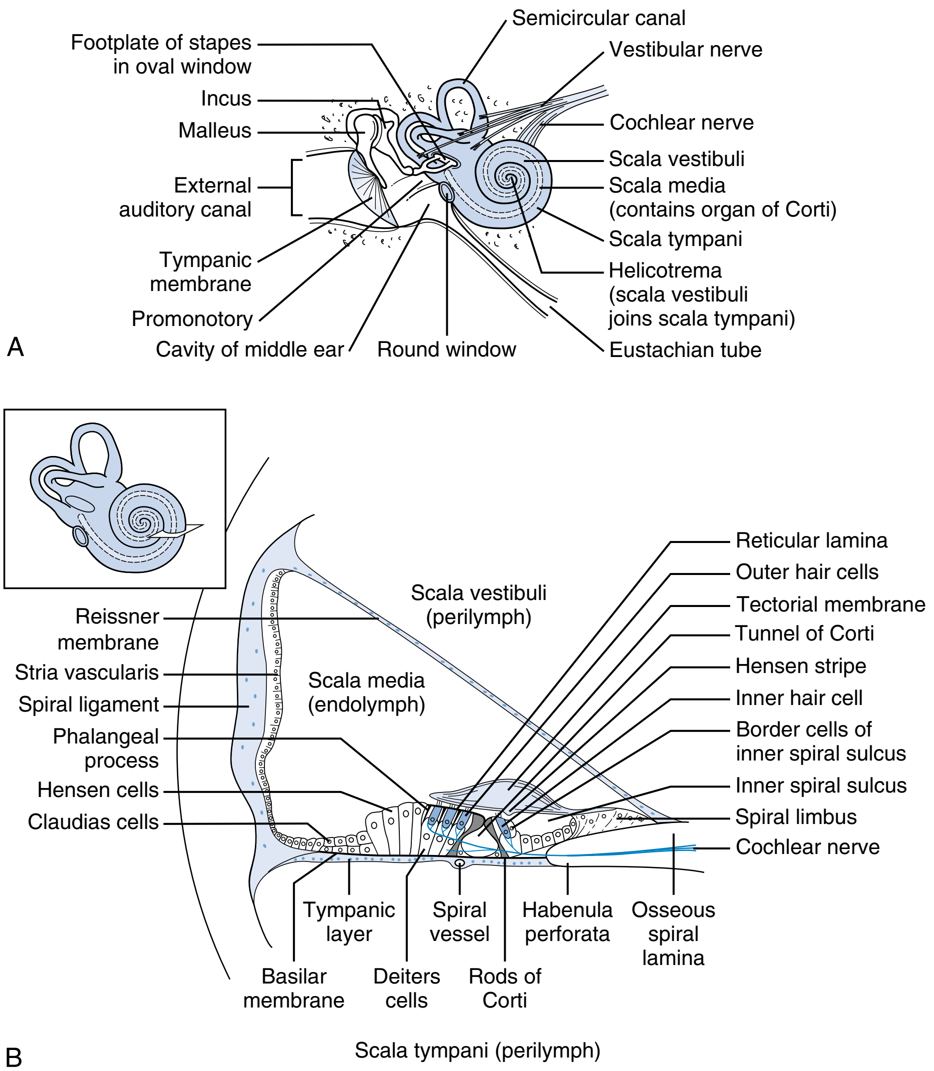
Hearing loss can result from dysfunction of and/or damage to the auditory system at any point, from the outer ear to the auditory cortex. Medical, environmental, lifestyle, and genetic factors are closely related to the etiology of hearing loss. If hearing impairments develop in early life, speech intelligibility, language, literacy, cognition, emotion, and social skills may be adversely affected. Elderly persons with significant hearing loss may have psychological impacts such as cognitive decline, anxiety, depression, lethargy, decreased independence, and social dissatisfaction.[62] Also important is the fact that hearing impairments can affect their communication with health care providers, possibly leading to the inappropriate management of serious health conditions.[167]

According to standard hearing examinations, approximately 13% of Americans age 12 years or older (i.e., approximately 30 million people) have bilateral hearing loss.[90] In general, hearing loss is likely to develop with advancing age as a consequence of a variety of factors, such as comorbid conditions (diabetes, cardiovascular disease, hyperlipidemia), lifestyle factors (diet, exercise, smoking), and exposure to high levels of noise and music. Males are more likely to experience age-related hearing loss than females. Disabling hearing loss was reported by 2% of the population in the 45- to 54-year-old range, increasing to 8.5% in the 55- to 64-year-old range, to 25% in the 65- to 74-year-old range, and to 50% in those aged 75 years and older.[104]

Overview of Auditory Anatomy and Physiology

While more comprehensive reviews of the anatomy and physiology of the auditory system exist (e.g., Hall[55]), in brief, sound is generated when air particles vibrate and propagate as sound waves. Our auditory system receives this air vibration and converts it to different energy forms (mechanical, hydrodynamic, and electrochemical) so that we perceive it as sound. Thus the ear is often described as an acoustic transducer (Video 4.1). Fig. 4.1A illustrates the anatomy of the ear. Sound from the environment travels through the external auditory canal to the middle ear. The middle ear consists of two major structures: the tympanic membrane (eardrum) and the ossicular chain, which is composed of the three ossicles: the malleus, incus, and stapes. The middle ear is an air-filled space that is connected to the posterior of the nasopharynx via the eustachian tube, which works to equalize air pressure between the middle ear and the outer environment. Sound vibrating the tympanic membrane sets the ossicular chain into motion. The middle ear amplifies sound as an impedance-matching transformer, thus preventing energy loss as sound travels from an air medium to a fluid medium (i.e., the inner ear).

The inner ear, a fluid-filled space within the temporal bone, consists of two parts: the cochlea part for hearing function and



• **Fig. 4.1** (A) Cross section of the inner ear in relation to the outer and middle ears. (B) Cross section of the organ of Corti in the scala media. (Redrawn from Yost WA. *Fundamentals of Hearing: An Introduction.* 3rd ed. Academic Press; 1994.)

the vestibular part for balance. The cochlea is divided into three divisions: the scala tympani, scala media, and scala vestibuli. Inner and outer hair cells, the sensory cells of hearing, rest on the basilar membrane, which separates the scala media and scala tympani (see Fig. 4.1B). The hair cells are mechanoreceptor cells that have important ion channels in their cilia (stereocilia) for electrochemical sound transduction. When the basilar membrane is displaced according to the vibration patterns of the oval window via the stapes, stereocilia bend because of pivotal motion of the basilar membrane and tectorial membrane. Bending of the stereocilia opens the hair cell's ion channels, leading to the release of neurotransmitters into the synaptic cleft between the base of the hair cells and the auditory nerve fibers to initiate action potentials.

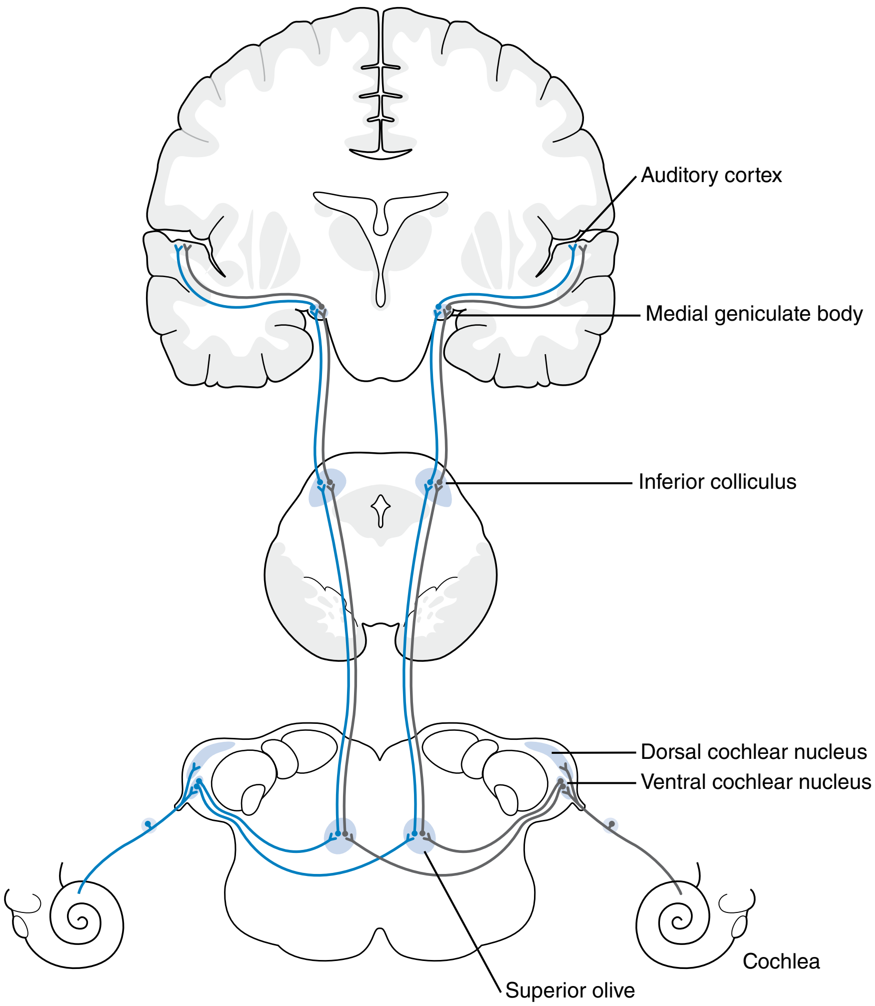
There are approximately 3500 inner hair cells in the human cochlea arranged in one row, in contrast to about 15,000 outer hair cells arranged in three rows. Most inner hair cells are innervated by afferent fibers at the base of the cells, encoding frequency information and contributing to sound clarity, whereas outer hair cells are innervated by mostly efferent fibers. Importantly, outer hair cells have the rather unusual property of motility. When activated by sound, they rapidly expand and contract to amplify the movement of the basilar membrane.

Outer hair cells contribute to the decoding of lower-intensity sounds and a high level of frequency specificity in hearing.

After the hair cells transduce mechanical energy into electrochemical activity, neural firing of the eighth cranial nerve generates action potentials that travel via brainstem nuclei and pathways and the thalamus to the auditory cortex located in the temporal lobe of the brain. Much auditory processing of complex speech signals in quiet and noise occurs in the centers of the central auditory nervous system. Fig. 4.2 shows the afferent auditory pathways for bilateral neuronal projection to the cochlear nucleus, superior olivary complex, and via lateral lemniscus to inferior colliculus. Auditory pathways continue bilaterally through the medial geniculate body in the thalamus to the auditory cortex in the temporal lobes of the brain.

Auditory Assessment

Various audiologic tests are performed to make a differential diagnosis of an individual's hearing status. Table 4.1 summarizes typical procedures used in hearing assessment and the sites evaluated with each test. Hearing screening can be performed by physicians, nurses, speech-language pathologists, audiologists, and technicians, often with automated devices. However, audiologists have



• **Fig. 4.2** The major components of the afferent auditory pathway. (Redrawn from Yost WA. *Fundamentals of Hearing: An Introduction.* 3rd ed. Academic Press; 1994.)

TABLE 4.1 Summary of Typical Audiologic Testing and Sites of Test

Audiologic Test	Type	Testing Site	What Is Tested
Otoscopic examination	Objective	Outer and middle ears	Structural integrity of the outer and part of the middle ear (tympanic membrane and part of the ossicular chain) based on visual inspection
Pure-tone audiometry	Subjective	Entire auditory system	Auditory thresholds through the air- and bone-conduction pathways at various frequencies based on behavioral response
Speech audiometry	Subjective	Entire auditory system	Ability to receive or discriminate speech based on behavioral response
Tympanometry	Objective	Outer and middle ears	Mobility of the tympanic membrane, function of the middle ear, ear canal volume, and obstruction of the ear canal
Acoustic reflexes	Objective	Retrocochlea	Function of the acoustic reflex pathway, including the stapedius muscles, auditory nerve, cochlear nucleus, superior olivary complex, and facial nerve
Otoacoustic emissions	Objective	Cochlea	Function of the outer hair cells
Auditory brainstem response	Objective	Retrocochlea	Function of the brainstem in response to sound

responsibility for conducting diagnostic hearing assessment as well as analyzing, interpreting, and reporting test results.

Case History

The history is an important part of hearing screening, diagnostic hearing assessment, and the diagnosis and management of hearing loss. Typical questions are listed in Box 4.1. Importantly, the history should include questions about the patient's medical and occupational history related to hearing and hearing loss. It is important to gather the history of noise exposure, ear surgery, ototoxic medication/chemical intake, and family history of hearing loss.

Otoscopic Examination

Inspection of the external ear and examination of the external ear canal and tympanic membrane with an otoscope is important to check for the presence of any structural abnormality. Box 4.2 lists abnormalities that must be considered during the ear inspection. Otoscopic examination should be conducted before hearing screening and diagnostic hearing testing because any obstruction of the ear canal can affect the results of pure-tone audiometry or other audiologic tests.

Test Battery for Auditory Assessment

Hearing assessment is optimally conducted with a test battery consisting of procedures requiring a behavioral response from the patient and with objective auditory procedures (see Hall[55] for review). Common behavioral tests include pure-tone audiometry and speech audiometry. The goal of pure-tone audiometry is an estimation of auditory thresholds for pure-tone signals presented via air conduction with earphones and via bone conduction with a small oscillator placed on the mastoid bone behind the ear. Pure-tone audiometry is a traditional hearing test that dates back to the 1920s. Testing is conducted in a very quiet setting, usually

• BOX 4.1 Typical Questions to Ask Patients During Case History Intake

- How long has the hearing difficulty been noticed?
- Was the onset of hearing loss sudden or gradual?
- Did the onset of hearing loss appear to have coincided with any event or trauma?
- In what sorts of situations is the hearing difficulty noticed?
- Does your hearing fluctuate within a day or from day to day?
- Is there any pain in the ear or drainage of it?
- Do you hear better in one ear than the other?
- Is there any pain in the ear or drainage out of it?
- Are there associated symptoms, such as tinnitus, dizziness, disequilibrium, headaches, or any other abnormality?
- Have you been diagnosed with any medical conditions, such as diabetes, cardiovascular disease, high cholesterol, arthritis, kidney disease, cognitive impairment, or visual impairment?
- Are you a smoker? If so, how long have you smoked, and much do you smoke?
- Have you ever been around loud noises or music?
- Have you ever had a hearing test before today? If so, what did the results indicate?
- Do you use or have you ever used any type of hearing aid or assistive device? If so, do you wear the device routinely and are you satisfied with it?

• BOX 4.2 Abnormalities That Must Be Checked for During the Otoscopic Examination

External Ear Structure

- Abnormalities (e.g., atresia, anotia, microtia)
- Set (position) of the ears
- Skin tags or sinuses, tenderness, redness or edema, signs of drainage or cerumen buildup

Internal Ear Structure

Ear Canal

- Obstruction (e.g., cerumen buildup, foreign objects)
- Signs of inflammation
- Drainage or blood
- Stenosis
- Damage

Tympanic Membrane (TM)

- Normal landmarks (cone of light, translucent/pearly gray TM, handle of malleus)
- Signs of inflammation (e.g., bright red TM, bulging TM)
- Retracted TM
- Perforated TM
- Bubbles behind TM

a specially built sound room. Patients are instructed to press a button or, less often, raise a hand whenever they just barely hear the tones.

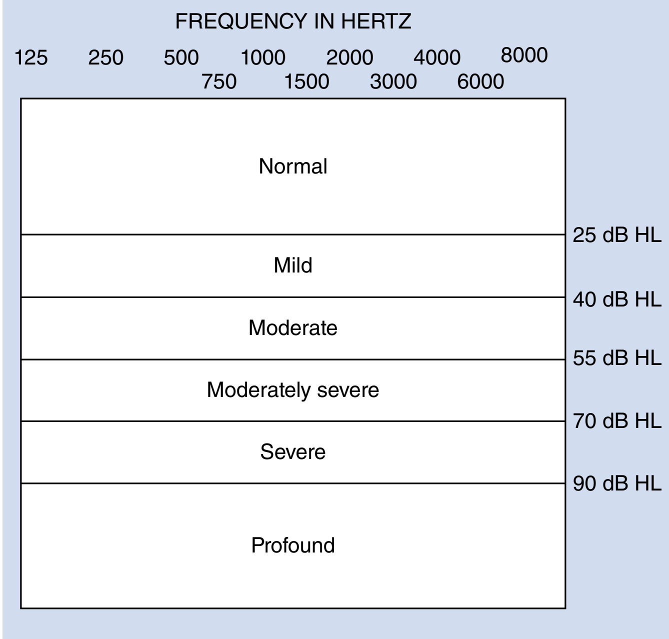
Pure-tone audiometry is invariably included in the test battery for the hearing assessment of older children and adults. However, it has some serious limitations as a clinical measure of auditory function, particularly in patients with cognitive and/or language impairment. In such patients, test performance is not always closely related to communication ability, and results are heavily influenced by listener variables such as motivation, attention, processing speed, among other cognitive factors.

Speech audiometry includes procedures for the detection and perception of speech signals such as words or sentences. For example, the patient is instructed to listen to a series of words and to repeat them as the intensity level is gradually reduced. Speech detection results and speech thresholds are reported as decibel (dB) levels. Speech audiometry also includes measurement of word recognition. Single-syllable words in a list of 25 words are presented to the patient via earphones at a fixed intensity level in a quiet sound room. Word recognition scores for each ear are calculated from the percentage of words that the patient repeats correctly. Speech audiometry performance as typically performed clinically may be compromised by listener factors, especially in young children, because of their cognitive and neurologic immaturity and in adult patients with traumatic brain injury (TBI). In addition, scores of clinically common tests of word recognition in quiet are not well correlated with a patient's ability to communicate effectively in typical listening conditions. That is, word recognition scores are often quite good in patients who have difficulty with speech perception in noisy environments. Furthermore, word recognition in quiet is an ineffective measure of auditory processing and is insensitive to central auditory nervous system abnormalities. Administration of a clinical test of speech perception in noise is a more appropriate option for such patients.

Types of hearing loss. Based on the results of a test battery of behavioral and objective auditory tests, hearing loss is confirmed and differentiated into its degree (Fig. 4.3) and into three types or categories: (1) conductive loss involving the middle ear, (2) sensorineural loss involving the inner ear or auditory nerve, and (3) mixed loss with involvement of the middle and inner ear.

Conductive hearing loss. Hearing loss caused by damage to the outer ear and/or middle ear is called conductive hearing loss. It is characterized by normal bone-conduction hearing thresholds and elevated air-conduction thresholds with an air-borne gap (difference between air-conduction and bone-conduction thresholds) of 15 dB or greater. This type of hearing loss can be caused by obstruction of the ear canal by foreign bodies or cerumen, outer or middle ear disorders, deformation of the outer and/or middle ears, mechanical injury to the outer and/or middle ears, and other middle ear disorders such as otitis media, otosclerosis, and cholesteatoma. In general, etiologies underlying conductive hearing losses are medically treatable. Tympanometry is abnormal in patients with conductive hearing loss.

Sensorineural hearing loss. Sensorineural hearing loss (SNHL) is characterized by elevated air-conduction thresholds with similar thresholds for bone-conduction pure-tone audiometry. Objective tests such as tympanometry confirm normal middle ear functioning and rule out conductive hearing loss.



• **Fig. 4.3** Classifications of the degree of hearing loss. *dB HL*, Decibels in hearing level.

SNHL is the result of damage to or dysfunction in the cochlea and/or the auditory nerve (cranial nerve [CN] VIII). Cochlear dysfunction is far more common than neural dysfunction, so the term *sensory hearing loss* is accurate in most cases. Both sound clarity and loudness are degraded in sensory or neural hearing loss. SNHL is permanent in most cases. Although it may be caused by various auditory disorders, such as tumors on CN VIII, Meniere disease (MD), and deformation of the cochlea and/or CN VIII, the most common etiologies—noise-induced hearing loss (NIHL), including acoustic trauma and age-related hearing loss—do not involve active disease or pathology. Other factors that contribute to the development of SNHL are ototoxic drugs or chemicals, hypoxia, TBI, infections (e.g., meningitis, rubella), immune system disorders, and genetics.

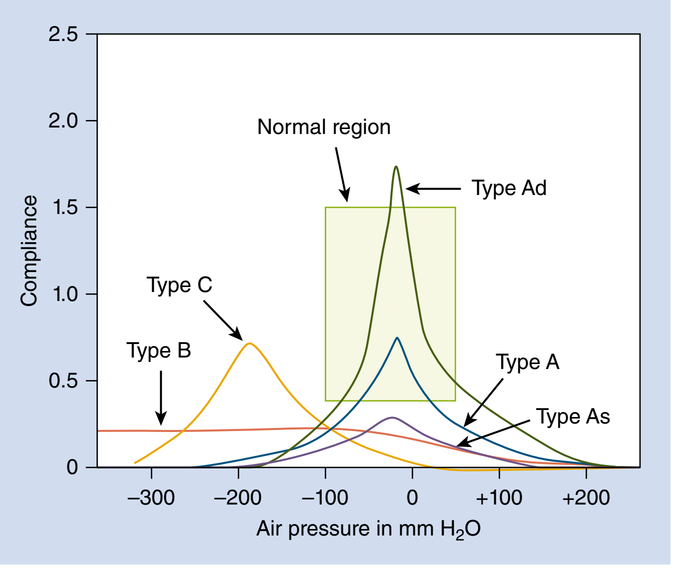
Mixed hearing loss. Mixed hearing loss, essentially a combination of conductive and sensory hearing loss, is characterized by elevated air-conduction and, to a lesser extent, bone-conduction thresholds.

Objective auditory tests. Objective tests are physiologic measures of auditory function that do not require a behavioral response from the patient. Valid objective auditory test results can be obtained from patients who are inattentive, sleeping, sedated, anesthetized, comatose, or unable for whatever reason to volunteer valid behavioral responses to sound. In other words, listener variables already noted have no effect on the outcome of objective auditory tests. A detailed description of these tests is far beyond the scope of this chapter. Readers interested in learning more about objective auditory measures are referred to a recent textbook (Hall[55]). There are also many hundreds of articles in the literature describing research evidence in support of the clinical application of each test.

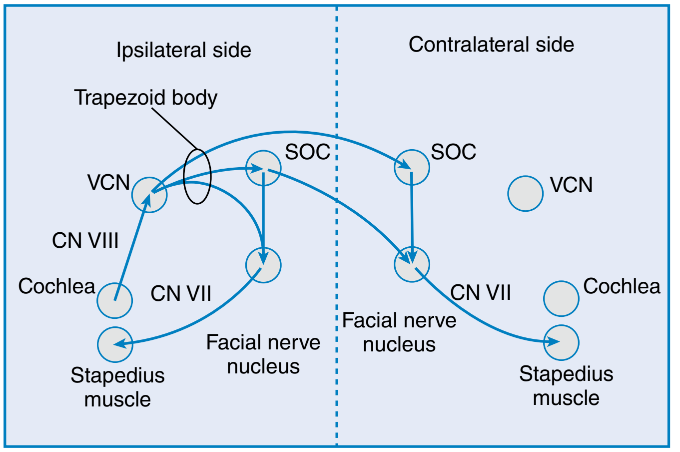
One objective auditory procedure is aural immittance measurement, which includes tympanometry and activation of the acoustic reflex. Tympanometry, performed with a soft rubber probe hermetically sealed within the external ear canal, records changes in compliance or stiffness of the middle ear as air pressure is varied in the external ear canal, usually from +200 daPa to −300 daPa. The resulting tympanogram is then analyzed and classified into one of several types (Fig. 4.4). Tympanometry is commonly included in the auditory assessment of young children, but it can be completed in 1 minute or less in patients of all ages. The resulting tympanogram provides detailed information on the function of the middle ear.

In acoustic reflex measurement, a loud sound (>75 to 80 dB) is presented to one ear while middle ear compliance or stiffness is monitored in the same ear or the opposite ear. A decrease in middle ear compliance, or increase in stiffness, occurs whenever the stapedius muscle contracts after the presentation of loud sounds. With a test time of less than 1 minute, acoustic reflex measurement provides objective information on the status of multiple regions of the auditory system, including the middle ear, cochlea, CN VIII, and brainstem pathways. Acoustic reflex pathways are illustrated in Fig. 4.5. Additionally, some information is obtained on function of CN VII (facial nerve), the efferent portion of acoustic reflex pathway that innervates the stapedius muscle.

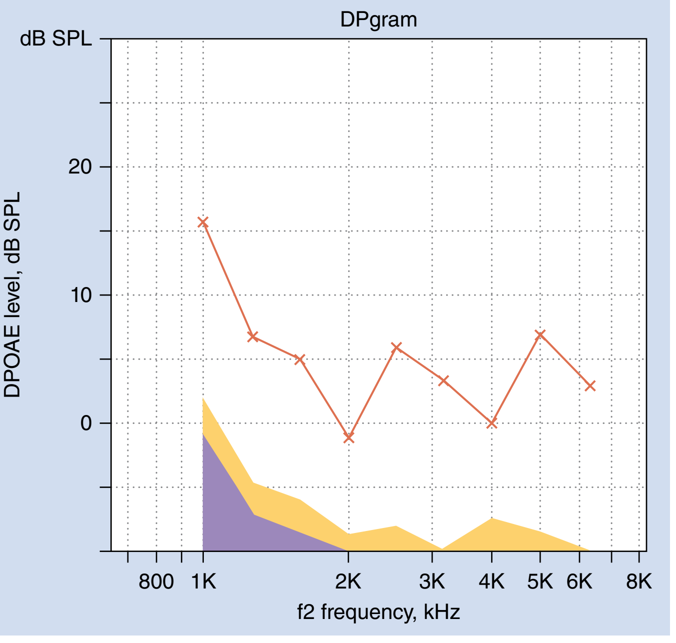
Otoacoustic emissions (OAEs) are very faint sounds recorded in the external ear canal that correspond to energy within the cochlea and are related to motility (up-and-down movement) of thousands of outer hair cells.[36] OAE measurement requires only 1 to 2 minutes for each ear. Stimulus sounds are presented via a small soft probe assembly inserted into the external ear canal. OAE-related sound emitted from the cochlea is detected with a miniature microphone housed within the probe. Outer hair cell function is invariably disrupted by most etiologies of cochlear damage, including age-related changes in older adults, exposure to high-intensity sounds and acoustic trauma, ototoxic drugs, and infections. Distortion product OAEs (DPOAEs) are the most common type applied clinically. DPOAE amplitude is displayed as a function of different test frequencies in a DPgram (Fig. 4.6).



• **Fig. 4.4** Classifications of the types of tympanograms. (Redrawn from Hall III JW, Mueller III HG. Audiologists' Desk Reference. Volume I: Diagnostic Audiology: Principles, Procedures, and Practices. Singular Publishing Group; 1997.)

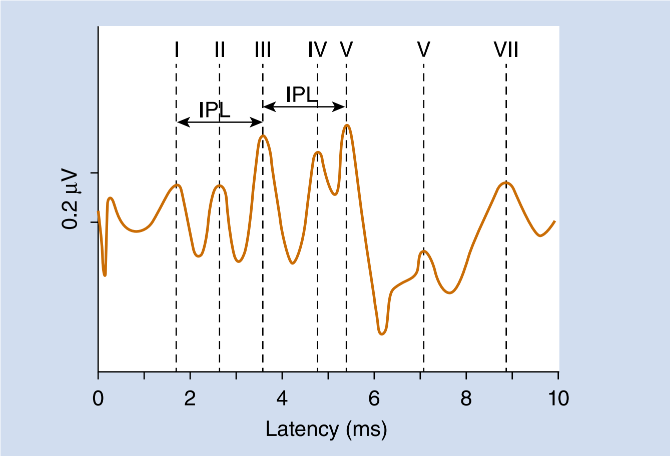


• **Fig. 4.5** Acoustic reflex pathway and anatomic structures. *CN VIII,* Auditory nerve; *CN VII,* motor branch of facial nerve; *SOC,* superior olivary complex; *VCN,* ventral cochlear nucleus. (Redrawn from Katz J (ed). The acoustic reflex. In *Handbook of Clinical Audiology*, ed 5. Philadelphia: Lippincott Williams & Wilkins, 2002.)



• **Fig. 4.6** DPgram. *dB SPL*, Decibels in sound pressure level; *DPOAE*, distortion product otoacoustic emissions.

Auditory-evoked responses reflect electrical activity from CN VIII as well as auditory pathways in the brainstem, thalamus, and cerebral cortex.[54] The responses are elicited with brief-duration sounds presented to each ear and detected with electrodes located on the scalp and near the ears. Auditory-evoked responses are electrophysiologic measures of the integrity of the auditory nervous system. One of them, the auditory brainstem response (ABR), is often applied clinically to objectively estimate frequency-specific hearing thresholds in patients who cannot participate in pure-tone audiometry, including infants and young children as well as adults with cognitive deficits after TBI. An ABR waveform is shown in Fig. 4.7.



• **Fig. 4.7** Schematic drawing of early latency response to an auditory stimulus of 60 dB at 10/s (brainstem auditory-evoked potential). *IPL*, Interpeak latency. (From Kowlgi NG, Khurana I, Khurana A. *Textbook of Medical Physiology.* 4th ed. Elsevier; 2024.)

Risk Factors for Hearing Impairments

Hearing impairments can be associated with a wide variety of etiologies, including genetic deficits/mutation, ear disease, environmental factors such as noise exposure or ototoxic drug intake, infections, and complications during pregnancy. Other factors such as age, gender, alcohol, smoking, diabetes, hypertension, stroke, cardiovascular diseases, and unhealthy lifestyle in general are also associated with hearing impairments.

Acquired Hearing Loss

Acquired hearing loss, developing any time after birth, is associated with various etiologies such as ear disease, viral (rubella, cytomegalovirus, mumps, acquired immunodeficiency syndrome, herpes) or bacterial infection (meningitis, syphilis), vascular insult to the cochlea, trauma to the auditory system, ototoxic drug/chemical intake, traumatic noise exposure, and aging. Some patients with acquired hearing loss have systemic disease, including thyroid disease, diabetes mellitus, kidney disease, multiple sclerosis, connective tissue disease, and neurofibromatosis type II.

Ototoxicity is one explanation for acquired hearing loss. Various drugs and chemicals are toxic to the auditory system.[22] The resulting sensory hearing loss is typically bilateral, permanent, sometimes progressive, and first detected in the high-frequency region. Development of hearing loss largely depends on types of medications, dosage, and duration of intake. Prior hearing loss and genetics also play a role in the likelihood of ototoxicity. Aminoglycoside antibiotics (e.g., amikacin, gentamicin, kanamycin, neomycin, netilmicin, streptomycin, tobramycin), loop diuretics, analgesics, antipyretics (e.g., high-dose aspirin, acetaminophen), antimalarial agents (e.g., quinine, chloroquine), and antineoplastic or chemotherapeutic agents (e.g., cisplatin) are known to cause either temporary or permanent sensory hearing loss. Lead, cadmium, mercury, manganese, styrene, toluene, ethyl benzene, and xylene are recognized as ototoxic chemicals. If patients are on ototoxic medication, clinical practice guidelines recommend that their hearing be monitored, and they should be asked to report any changes in hearing.

Noise-Induced Hearing Loss

Acute traumatic and chronic noise exposure can cause mechanical damage and/or chemical reactions called oxidative stress to the auditory system,[63] leading to either temporary or permanent sensory hearing loss. NIHL is the most common acquired hearing loss and a major source of sensory impairment in civilian and military populations. The audiologic profile of typical NIHL is characterized by bilateral, high-frequency hearing loss with a notch (greater hearing loss) in the audiogram within the frequency region of 3000, 4000, or 6000 Hz. However, gunfire can cause an asymmetric sensory hearing loss greater in the ear closest to the side of the gun or weapon. Tinnitus often accompanies NIHL. The four major aspects of noise that contributes to development of hearing impairments are (1) overall noise level, (2) spectrum distribution of noise, (3) duration and distribution of noise, and (4) cumulative noise exposure in days, weeks, or years.

Age-Related Hearing Loss

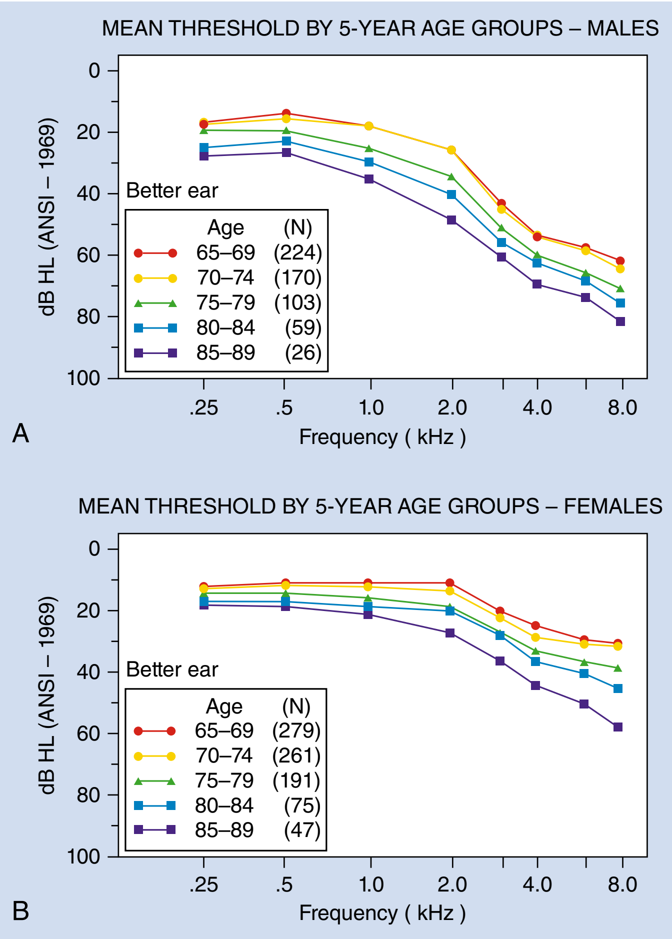
Age-related hearing loss, also referred to as presbycusis, is the third most prevalent chronic medical condition in the geriatric population. More than 10 million Americans over 65 years of age exhibit hearing impairment. Hearing loss with advancing age is a complex disorder with contributions from genetic and nongenetic (environmental, lifestyle, medical) factors.[156] Multiple structures within the inner ear are involved in age-related hearing loss. This type of hearing loss is sensorineural (usually sensory), permanent, and bilateral. High-frequency hearing is affected more than low-frequency hearing (Fig. 4.8).

Auditory Processing Disorder

A patient with auditory processing disorder (APD) exhibits difficulties processing auditory information, often without clinically significant peripheral hearing loss.[55] Some common characteristics are difficulty understanding speech in the presence of noise, inability to understand degraded or rapid speech, and difficulty localizing the source of a signal. APD is seen in patients with TBI, in children (with an apparent neurodevelopmental or neurobiologic explanation), and typically in aging adults. There are now effective management strategies for improving communication in persons with APD, assuming that the problem has been identified and accurately diagnosed.

Tinnitus

Tinnitus is a symptom, not a disease. Bothersome tinnitus is a disorder most often associated with inner ear dysfunction. Tinnitus does not cause hearing loss but is often reported by persons with hearing loss. Tinnitus is the perception of sound in the absence of external sound sources; it is a phantom auditory perception. A wide variety of different sounds are perceived as tinnitus, although a high-pitched complex ringing sound is most common. Pulsatile or intermittent tinnitus may be related to a serious health condition such as vascular disease or tumors. A substantial proportion of persons with bothersome tinnitus, about 10% of the general population, experience disruption of concentration and/or sleep, which is debilitating.[10] Chronic bothersome tinnitus has a major impact on quality of life and may lead to clinical depression. Bothersome tinnitus is most often associated with age-related or noise-induced auditory dysfunction, not ear disease or pathology. A substantial research effort in the past decade has provided clinically useful information on mechanisms of tinnitus, how the perception of bothersome



• **Fig. 4.8** Changes in hearing thresholds in different age categories. (A) Males. (B) Females. *ANSI,* American National Standards Institute. (Redrawn from Gates GA, Cooper Jr JC, Kannel WB, et al. Hearing in the elderly: the Framingham cohort, 1983–1985 Part I. Basic audiometric test results. *Ear Hear.* 1990;11:247-256.)

tinnitus is represented in the nervous system, and why tinnitus is so distressing for some patients. Evidence-based audiologic options for the management of tinnitus include diet and lifestyle modifications, hearing aids, sound therapy to desensitize the perception of tinnitus, counseling, and tinnitus retraining therapy.[55]

Red Flags: Warning of Ear Disease

The American Academy of Otolaryngology–Head and Neck Surgery has identified 10 red flags or warnings of ear disease (Box 4.3). Patients who show any signs of red flags should be referred to a physician and/or specialist (i.e., otolaryngologist, otologist) for further evaluation and treatment. A recently reported 15-item questionnaire—the Consumer Ear Disease Risk Assessment (CEDRA)—is another option for identifying patients who are at risk for ear disease.[108] Available online (https://sites.northwestern.edu/cedra/), the CEDRA was developed with funding from the National Institutes of Health to sort out the relatively small proportion of adults with ear disease who should seek medical care from the majority of patients with hearing loss secondary to noise exposure or aging that is not associated with disease or pathology.

• BOX 4.3 Ten Red Flags[a]: Warning of Ear Disease

1. Hearing loss with a positive history of ear infections; noise exposure; familial hearing loss; tuberculosis; syphilis; human immunodeficiency virus; Meniere disease; autoimmune disorder; ototoxic medication use; otosclerosis; von Recklinghausen neurofibromatosis; Paget disease of bone; ear or head trauma related to onset
2. History of pain, active drainage, or bleeding from an ear
3. Sudden onset or rapidly progressive hearing loss
4. Acute, chronic, or recurrent episodes of dizziness
5. Evidence of congenital or traumatic deformity of the ear
6. Visualization of blood, pus, cerumen plug, foreign body, or other material in the ear canal
7. An unexplained conductive hearing loss or abnormal tympanogram
8. Unilateral or asymmetric hearing loss (a difference of >15 dB pure-tone average between ears) or bilateral hearing loss >30 dB
9. Unilateral or pulsatile tinnitus
10. Unilateral or asymmetrically poor speech discrimination scores (a difference of >15% between ears) or bilateral speech discrimination scores <80%

[a]The red flags do not include all indications for a medical referral and are not intended to replace clinical judgment in determining the need for consultation with an otolaryngologist.

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Symptoms of Hearing Loss

It is very common for patients to be unaware of the presence of hearing impairments. Hearing loss in adults is most often not associated with illness or disease, and it typically progresses slowly. As a result, adults with hearing loss may deny any problem or express indifference about their condition. Older patients with hearing loss may remain in a lengthy denial stage because they view hearing loss as an inevitable result of aging. Stigma of wearing a hearing aid adds to long delay in seeking hearing health care. Often, family members or friends may complain that the patient does not follow conversation or listens to the television with the volume turned up too high. Box 4.4 lists typical complaints related to hearing impairment.

It is very important for health care providers to conduct hearing screening for the early detection of hearing impairment. Screening for hearing impairment with validated questionnaires or inventories such as the Hearing Handicap Inventory for the Elderly–Screening version[156] and Hearing Handicap Inventory for Adults[106] can be quickly and easily performed in a physician's office to detect patients with communicatively important hearing loss. Automated

• BOX 4.4 Typical Complaints of Patients with Impaired Hearing[a]

- Difficulty understanding speech in background noise (e.g., restaurant)
- Difficulty understanding speech in a group setting (e.g., meeting)
- Able to hear speech but unable understand what is said
- The impression that people are mumbling
- Speaking too loudly (sensorineural hearing loss) or too softly (conductive hearing loss)
- Turning up volume on television and/or difficulty using a phone
- Difficulty understanding accented and/or unclear speech
- Difficulty learning a foreign language
- Tinnitus and/or dizziness
- Difficulty following a conversation
- Complaints from family members or friend about the patient's hearing loss

[a]Some people may have no symptoms at all if hearing loss progresses slowly overtime.

• BOX 4.5 Patient's Signs That the Physician Should Note Regarding Possible Hearing Impairment

- Cannot hear or understand you when you ask the patient to enter your office for the appointment
- Not very talkative
- Does not reply when spoken to
- Does not respond appropriately to questions you are asking
- Does not understand you when you speak on the telephone
- Looks closely at your face or lips when you speak
- May seem confused when giving an inappropriate response
- May have a hearing aid but does not use it
- May strain to understand or place a hand behind the ear in an attempt to amplify your voice
- Needs occasional repetition or a louder speaking voice to understand
- Needs frequent prompting, multiple repetitions, or restatement of what was said
- Is unable to understand what is being said; asks you to write the message down to facilitate understanding
- Appears to have difficulty understanding fast speech and accented speech

From Weinstein BE. *Geriatric Audiology.* 2nd ed. Thieme Medical Publishers; 2013.

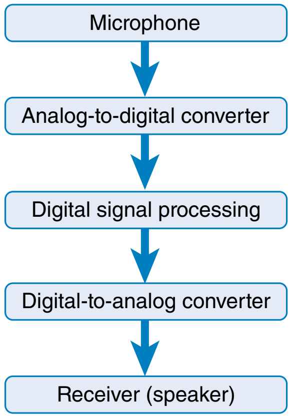
OAE devices also offer a clinically feasible approach for routine hearing screening in children and adults with any risk for hearing loss. Box 4.5 lists signs that physicians should note regarding possible hearing impairments. Another important issue to keep in mind is that cognitive impairment may masquerade as hearing loss or, in the elderly population, may coexist with hearing loss, thus further compromising everyday communication.

Auditory Rehabilitation

Auditory *rehabilitation* is a term designating strategies and techniques for restoring or optimizing hearing and listening abilities as well as effective communication. Auditory *habilitation*, on the other hand, designates services to children (<18 years of age) with congenital hearing loss or hearing loss present at birth or acquired before the development of speech and language. Audiologists, speech-language pathologists, and sometimes otolaryngologists work together to provide auditory rehabilitative/habilitative services to individuals with hearing loss. Auditory rehabilitation includes technology (e.g., hearing aids, hearing assistance technology, cochlear implants [CIs]) and auditory training.

Hearing Aids

Hearing aids are electrical devices that provide amplification of sound specifically for an individual's degree and configuration of hearing loss. Digital hearing aids are the most common form of noninvasive and nonmedical management for improving speech detection and perception in persons with permanent hearing loss. Space does not permit a detailed description of hearing aids. Briefly, the main components in digital hearing aids are (1) a microphone, (2) an analog-to-digital converter, (3) a digital signal processing circuit, (4) a digital-to-analog converter, (5) a receiver (speaker), (6) a battery, and (7) a means of coupling the amplified sound into the ear canal (Fig. 4.9). If a patient has a reasonably symmetric bilateral hearing loss, two hearing aids are almost always recommended to maximize natural binaural hearing.



• Fig. 4.9 Simple diagram of hearing aids.

Various styles and types of hearing aids are illustrated in Fig. 4.10. Once hearing aids are selected and ordered, patients are scheduled for fitting and orientation sessions. An audiologist programs digital hearing aids to provide appropriate amplification for a specific person. For successful use, it is critically important to educate new hearing aid users on appropriate expectations. Some time is required for a new hearing aid user to make the most of amplified sound and, through neuroplasticity, to obtain optimal communicative benefit from amplification. To communicate more effectively, some persons may require hearing assistance technology in addition to traditional hearing aids. For example, frequency modulation (FM) is hearing assistive technology that transmits a speaker's voice from a remote microphone directly to the ears of the person with a hearing loss to maximize the difference between the speech signal and background noise. FM technology improves communication abilities in persons with APDs and normal hearing sensitivity as well as in some persons with hearing loss. A personal FM system can also be of great help in a physician's office to aid communication with a patient who is hearing impaired.

Surgical Management With Implantable Devices

Cochlear Implants

CIs are devices that electrically stimulate auditory nerve fibers via electrodes surgically implanted in the cochlea, thus bypassing functions of the outer, middle, and inner ear. A CI is composed of an external speech processor, transmitting coil, internal receiver/simulator, and electrode array. The patient wears the external processor, which picks up acoustic sound via a microphone and converts it to an electrical signal. The electrical signal is then converted to electromagnetic energy, which is transmitted from an induction coil to the internal receiver/simulator embedded subcutaneously on the skull. The internal receiver/simulator converts the signal to electrical pulses and sends it to the electrode array that is inserted by an otologist into the scala tympani duct of the cochlea. After the surgery, an audiologist programs the CI's speech processor based on objective and subjective measurements. A person with a CI requires auditory training because auditory perception with a CI is very different from what normal hearers or hearing aid users perceive.

• **Fig. 4.10** Different styles of hearing aids. (A) Completely-in-the-canal style. (B) Receiver-in-canal style. (C) Thin open style. (D) Receiver-in-canal behind-the-ear style. (Courtesy Widex USA Inc., Long Island City, NY.)

Bone-Anchored Hearing Aids

A bone-anchored hearing aid (BAHA) is a surgically implanted device that delivers sound via bone vibration to the cochlea. Individuals with conductive or mixed hearing loss, single-sided deafness, and an inability to tolerate conventional hearing aids in the ear canal are good candidates for the BAHA. In such a case, the patient has a small metal post—to which a sound processer is attached—implanted in the mastoid bone. The processor converts incoming sound to vibrations that are conducted to the inner ear via the skull, bypassing the outer and middle ear.

Middle Ear Implantable Hearing Aids

Middle ear implantable hearing aids are designed to treat conductive hearing loss and SNHL. The implanted portion of the device is attached to the ossicular chain and converts acoustic energy to mechanical vibratory energy. There are two types of middle ear implants: piezoelectric and electromagnetic. Candidates are patients who do not benefit from or cannot use conventional hearing aids (e.g., allergies/sensitivity to ear mold materials, chronic external ear infection) but have hearing loss that is not severe enough to qualify for a CI.

CASE 4.1

Auditory Impairment

A 75-year-old male, a recently retired Vietnam-era war veteran, presented to the audiology service with two chief complaints: (1) considerable difficulty hearing people speak, especially females in noisy settings, and (2) constant perception of a high-pitched ringing-type sound that interferes with his daily activities, including sleep. A detailed history revealed service-related exposure to high levels of sound in the 1960s and long-term recreational noise exposure (e.g., hunting with a rifle, attendance at stock-car auto races, motorcycle riding). The history was negative for head trauma and ear disease. The patient reported a progressive worsening of his ability to communicate, which had contributed to his early retirement (he had been unable to perform his duties as an automobile salesman). The patient's spouse confirmed on the day of the assessment that his hearing problems and annoyance with the ringing sounds were putting increasing stress on their marriage.

Otoscopy showed a moderate amount of cerumen in each external ear canal, but the tympanic membranes appeared normal. Inspection of the external ear (pinna) showed several raised skin discolorations. Tympanometry confirmed normal middle ear function. Acoustic reflexes were not measured because of the patient's complaint of bothersome tinnitus and reduced

CASE 4.1—cont'd

Auditory Impairment

tolerance to loud sounds. OAE measurements were abnormal and consistent with cochlear (outer hair cell) dysfunction for test frequencies above 1000 Hz. Pure-tone audiometry (air and bone conduction) revealed a moderate-severe (40 to 65 dB HL) symmetric, bilateral, high-frequency SNHL. Word recognition in quiet was poor (60 to 70%) for each ear at a high-intensity level of 80 dB HL and very poor (<40%) at a typical conversational listening level of 50 dB HL. Speech perception in noise (Hearing in Noise Test) showed abnormally low performance with a favorable signal (speech) to noise ratio of +5 dB HL.

The following management steps were taken based on these audiologic findings: (1) counseling of patient and spouse that included an explanation of the test results, the importance of a healthy lifestyle (no smoking, healthy diet, physical exercise), his likely benefit from amplification (hearing aids with remote microphone technology) for communicating in quiet and noisy settings and for reducing his perception of bothersome tinnitus, and the importance of hearing protection during an exposure to high-intensity sounds; (2) referral for formal tinnitus assessment and management with progressive tinnitus therapy; (3) otolaryngology consultation to rule out ear disease and to confirm age-related and noise-induced etiologies for the hearing loss; and (4) the recommendation of a follow-up visit to the audiology clinic in 6 months to monitor the patient's compliance and progress with these recommendations.

Vestibular Impairments

The vestibular neurons of CN VIII project centrally to both the cerebellum and the vestibular nuclei, where they synapse and transmit afferent vestibular inputs and contribute to vestibular reflex pathways. The vestibular end organs consist of the semicircular canal (SCC) system and otolith organ system. Both the peripheral and central vestibular systems contribute to the functions of balance, equilibrium, orientation in space, ocular stability during movement, and maintenance of posture. The visual and somatosensory systems also have conjunctive contributions to many of the previously mentioned functions. Vestibular system dysfunction can lead to a multitude of problems, including dizziness, vertigo (illusion of motion), disequilibrium, and falls.

The National Institute on Deafness and Other Communication Disorders[103] has reported that approximately 15% (almost 33 million) of American adults have complained about problems with dizziness or impaired balance. In older adults, balance disorders are related to falling, and the cost of emergency department visits associated with fall-related injuries are estimated to be $4 billion annually.

Additionally, large numbers of current or former US service members have sustained some form of TBI while serving in the military. Mild TBI related to blast exposure has become one of the most prevalent injuries documented among post-9/11 US veterans. The Defense and Veterans Brain Injury Center[34] has reported that the Defense Medical Surveillance System and the Armed Forces Health Surveillance Center tallied more than 380,000 TBIs in US military service personnel from 2000 through the first quarter of 2018. Military personnel with mild TBI are significantly more likely to have vestibular and balance/postural disorders, and in this population vestibular and oculomotor deficits tend to be more persistent than cognitive impairments.[143] Undetected vestibular disorders can lead to hastened return to duty and be detrimental to the success of military operations.

Overview of Vestibular Anatomy and Physiology

CN VIII is also called the vestibulocochlear nerve because it has two components: the cochlear nerve, which is responsible for hearing, and the vestibular nerve, which is related to balance and equilibrium. Fig. 4.11 illustrates the vestibular end organs and their orientation within the head. The three nearly orthogonally positioned SCCs (posterior, horizontal/lateral, anterior/superior) sense rotational/angular acceleration (see www.youtube.com/watch?v=YMIMvBa8XGs). The two otolith organs (utricle and saccule) detect linear acceleration, head tilt, and gravity. The stereocilia of the sensory hair cells within the utricle and saccule are embedded in a gelatinous structure (otolithic membrane) with associated microscopic calcium carbonate crystals (otoconia). The abnormal migration of otoconia into the SCC system results in a type of dizziness termed benign paroxysmal positional vertigo (BPPV).

There are three main vestibular reflexes: the vestibulospinal reflex (for body stabilization), vestibulocollic reflex (for head stabilization), and vestibuloocular reflex (VOR) (for visual gaze stabilization). The VOR, for example, stabilizes eye movement with respect to the visual field during head motion. Many test procedures used to evaluate the vestibular system require monitoring of eye motion related to vestibular function. The presence of nystagmus (involuntary eye movements) can be related to vestibular dysfunction. Jerk nystagmus consists of eye drift in one direction (slow phase) and a quick corrective eye movement (fast phase). These involuntary eye movements can occur in a side-to-side (horizontal nystagmus), up-and-down (vertical nystagmus), or circular (torsional or rotary nystagmus) pattern.

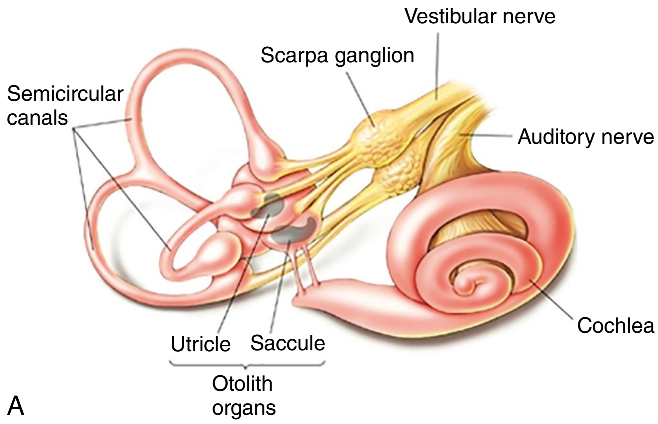
Kheradmand et al.[81] provided further insight into the perception of spatial orientation. By means of transcranial magnetic stimulation, they found that the posterior aspect of the supramarginal gyrus is involved in processing information from different sensory modalities into an accurate perception of being in an upright position.

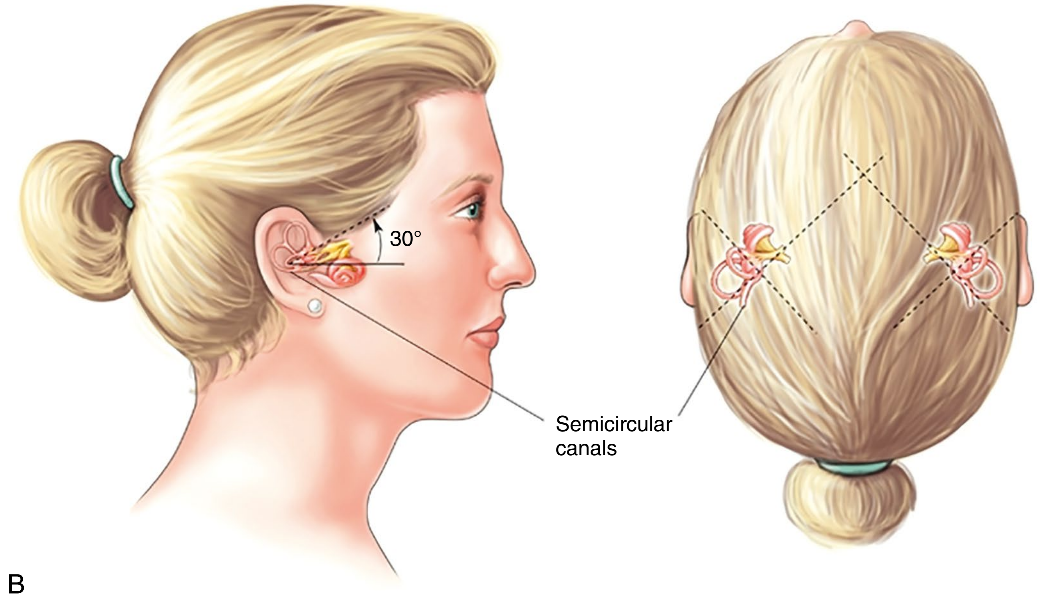
Vestibular Symptoms

Typical complaints from patients with vestibular impairments include dizziness, vertigo, and imbalance; they can be intermittent, episodic, or persistent. A variety of vestibular symptoms can be further defined during history taking. Symptoms can range from mild to severe. These vestibular symptoms can be spontaneous or may be activated by specific triggers such as body/head position and visual/sound stimuli. Moreover, different types of nystagmus (rapid involuntary eye movements) may be observed. Key vestibular symptoms are categorized into four types (dizziness, vertigo, vestibulovisual symptoms, and postural instability) with their subtypes; according to the Bárány Society, these categories are recommended for use in documenting patients' descriptions of their symptoms.[15]

Vestibular Assessment

In general, assessment of vestibular impairments comprises history taking, administration of questionnaires, physical examination,





• **Fig. 4.11** Vestibular end organ and orientation. (A) Schematic drawing of the cochlea and the peripheral vestibular apparatus with innervation from the auditory nerve and the vestibular nerves. (B) Orientation of the semicircular canals in the vestibular apparatus. (From Bear MF, Connors BW, Paradiso MA. The vestibular labyrinth. In *Neuroscience: Exploring the Brain.* Lippincott Williams & Wilkins; 2007.)

computerized vestibular testing, and radiology testing, including targeted computed tomography and magnetic resonance imaging. A thorough history is paramount in determining the cause of dizziness. A differential diagnosis is solidified with the qualitative information gathered from the questions. Further testing of the vestibular system can follow if quantifiable information is needed to confirm a diagnosis.

The foundation for the history taking is established with a thorough dizziness questionnaire. Gathered information should include the patient's description of the symptoms, the duration of the episode or episodes, and concomitant medical problems. Bennett[14] presented the following questions for use as the foundation of a dizziness history: "*Question 1:* Describe what you are experiencing. *Question 2:* How long does your dizziness last? *Question 3:* How often do you have attacks of vertigo? *Question 4:* Is there anything you can do that will cause you to feel dizzy (precipitating factors)? *Question 5:* What other symptoms do you have around the time of vertigo attacks (associated symptoms)? *Question 6:* Do you have any other medical problems? *Question 7:* What medications are you currently taking?"

Several questionnaires are available that can be used to assess subjective adjustment in patients with vestibular disorders. The Dizziness Handicap Inventory (DHI) is a commonly used subjective scale used in vestibular rehabilitation and was developed in an effort to quantify the handicapping effects of dizziness on a patient's well-being.[72] The original DHI consists of 25 items separated into three categories: functional, physical, and emotional. Whitney et al.[162] found that DHI scores above 60 are associated with severe dizziness and recurrent falls. The DHI can also be used to determine improvement after rehabilitation. Jacobson and Newman[72] found that a decrease on the DHI of 18 can be considered clinically significant progress.

The Activities-Specific Balance Confidence (ABC) is another questionnaire that can be used to assess patients' confidence in their balance function while performing everyday activities in the home and community.[121] The ABC can aid in the identification of patients' lack of confidence in performing common functional undertakings and their potential restriction of activities. It has also been used to document change over the course of rehabilitation.[161]

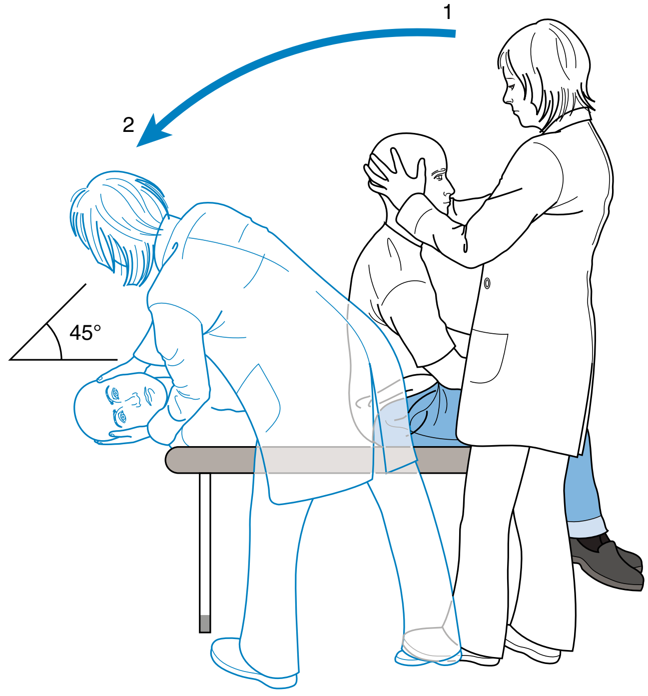
Following the history and administration of questionnaires, physical examination is usually performed. Some of the more common physical assessments of the vestibular system may include visualization of gaze-evoked nystagmus, the Romberg test, the Fukuda stepping test, and the Dix-Hallpike maneuver. Gaze testing involves monitoring for the presence of nystagmus or other abnormal eye movements while the patient fixates on stationary visual targets. Gaze nystagmus is considered abnormal if it is present with appropriately positioned visual targets. It can be interpreted with the Alexander law, which states that gazing in the direction of the fast phase enhances the nystagmus.

Jacobson et al.[71] assessed the sensitivity, specificity, and positive and negative predictive values of the Romberg test in patients with vestibular system lesions affecting the horizontal SCCs, saccule, and/or inferior and superior vestibular nerves. The Romberg test was compared with the cervical vestibular-evoked myogenic potential (VEMP) and caloric tests (see later). They concluded that the performance characteristics of the Romberg test were poor and insufficient for use as a screening measure for vestibular hypofunction.

The Fukuda stepping test assesses labyrinthine function through the vestibulospinal reflexes. The patient steps in place on a marked grid with eyes closed and arms stretched out forward. Measurements of rotation and displacement are taken after 50 to 100 steps and compared with data from healthy control subjects. Honaker et al.,[68] however, caution that the Fukuda stepping test may not be an accurate screening tool for peripheral vestibular asymmetry in chronically dizzy patients.

The Dix-Hallpike maneuver is very often used to assess for the effects of displaced otoconia in the posterior SCC. The abnormal migration of otoconia into the SCC system results in BPPV, which is considered to be the most common vestibular ailment and causes approximately 50% of dizziness in older people. In the Dix-Hallpike maneuver, the patient is seated on the examination bed with the head rotated to the lesion side by 45 degrees. The patient is then quickly lowered backward to a supine position, with the head hanging over the edge of the bed by 30 degrees. This places the patient in a position that aligns the posterior SCC in a vertical orientation. Potential debris in the posterior SCC would be influenced by gravity and thus cause abnormal fluid movement within the affected canal. The patient's eyes are open throughout testing while the examiner observes for the presence of nystagmus. Nystagmus with a predominant torsional component is present in classic BPPV. If nystagmus appears, the position is maintained until the nystagmus terminates before quickly returning the patient to an upright position. The signs and symptoms of BPPV are fatigable; therefore, if nystagmus and vertigo are present, the maneuver should be repeated immediately to evaluate the response for fatigue. Fig. 4.12 illustrates the maneuver for evaluating right posterior SCC BPPV.

In addition to the history, questionnaires, and physical examination, a variety of sophisticated computerized vestibular tests are available for those cases of dizziness in which the history and physical examination alone are inadequate or when there is a need for further corroborative data to confirm a diagnosis. These tests can provide information related to the site of dysfunction (peripheral, central, or both) and can be applied after treatment to provide information regarding the potential functional benefits of vestibular rehabilitation. Typical targeted vestibular testing includes electronystagmography/videonystagmography, rotational chair testing, subjective visual vertical testing, video head impulse testing, VEMP, computed dynamic visual acuity, the Gaze Stability Test,



• **Fig. 4.12** Right-side Dix-Hallpike maneuver. (Redrawn from Fife TDI, Lempert T, Furman JM, et al. Practice parameter: therapies for benign paroxysmal positional vertigo [an evidence-based review]: report of the Quality Standards Subcommittee of the American Academy of Neurology. *Neurology.* 2008;70:2067-2074.)

and computed dynamic posturography. Figs. 4.13 and 4.14 show some examples of targeted vestibular testing. Further information on computed vestibular testing may be obtained from free webinars available at the Interacoustics Academy website (https://www.interacoustics.com/academy/webinars) or a comprehensive textbook.[73] Furthermore, it is important to perform audiometric and visual assessments in conjunction with vestibular evaluation to rule out dual or multiple sensory impairments.

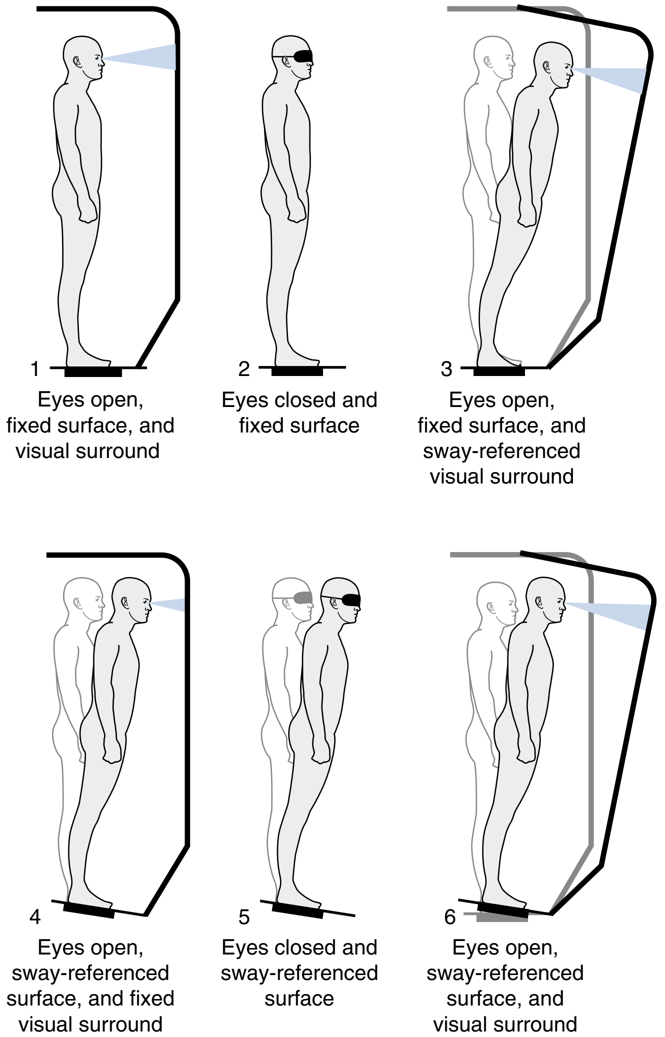
Risk Factors, Comorbidities, and the Epidemiology of Vestibular Disorders

The more common vestibular disorders include BPPV, vestibular neuritis, vestibular migraine, and MD. In Baloh et al.'s study on BPPV,[12] the mean age of onset was in the fifth decade of life, and almost half of the cases were idiopathic. Cases with a probable underlying cause included head trauma and viral neurolabyrinthitis. In an observational study on the role of comorbidities, De Stefano et al.[33] found a statistical link between the presence of systemic disorders (hypertension, diabetes, osteoarthrosis, osteoporosis, and depression) and BPPV.

Vestibular neuritis was the second most common cause of dizziness after BPPV identified in general practice clinics.[56] Adamec et al.[1] identified the most common comorbidities in patients with vestibular neuritis in their practice as hypertension, diabetes, hyperlipidemia, and hypothyreosis. Chuang et al.[26] suggested that comorbid vertebral artery hypoplasia may be a risk factor for severe vestibular neuritis.

Epidemiologic studies have shown a relationship between migraine and vertigo.[85] Neuhauser et al.[105] found the prevalence of

• **Fig. 4.13** Rotational chair. (From Bahner C, Petrak M, Beck DL, et al. In the trenches, part 3: caloric and rotational chair tests. *Hear Rev.* 2013;20. www.hearingreview.com/2013/03/in-the-trenches-part-3-caloric-and-rotational-chair-tests/. Courtesy Interacoustics A/S, Assens, Denmark.)



• **Fig. 4.14** Six sensory conditions for the Sensory Organization Test in computed dynamic posturography. *1,* Eyes open, fixed surface, and visual surround. *2,* Eyes closed and fixed surface. *3,* Eyes open, fixed surface, and sway-referenced visual surround. *4,* Eyes open, sway-referenced surface, and fixed visual surround. *5,* Eyes closed and sway-referenced surface. *6,* Eyes open, sway-referenced surface, and visual surround. (Courtesy Natus Medical Incorporated.)

vestibular migraine to be 7% in patients from a dizziness clinic and 9% in patients from a migraine clinic. Vestibular migraine was found to occur at any age and has a strong female preponderance.[37] Familial occurrence was identified as a risk factor.[110] Uneri's[155] retrospective study found that migraines were three times more common in patients with confirmed BPPV than in the general population. MD was also found to have an increased prevalence in migraineurs.[126]

MD is characterized by recurring episodes of long-lasting vertigo, fluctuating SNHL, tinnitus, and aural fullness. Its prevalence is difficult to ascertain because of its fluctuant nature and the potential absence of certain signs and symptoms in the early stages.[137] However, the estimated prevalence in the United States is 0.19%, with a dramatic increase in prevalence with age.[4] Genetic predisposition has been identified. Arweiler et al.[8] detected an autosomal dominant hereditary pattern and proposed a multifactorial etiology for MD.

In addition, head trauma resulting from motor vehicle accidents, sports-related injuries, and military training/deployment are known to cause vestibular dysfunction.[44,149] A variety of vestibular symptoms are reported in military service members, veterans, and civilians, including professional and amateur athletes.

Vestibular Rehabilitation

Medical treatment options for vestibular dysfunction include pharmacologic intervention and surgical treatment for disabling dizziness and vertigo. Other management options include vestibular rehabilitation,[64] physical therapy, and psychological intervention. These treatments are used alone or in combination depending on patient need and the diagnosis.

Vestibular rehabilitation is an exercise-based therapy that enhances central compensation, leading to improvements in symptoms such as dizziness, vertigo imbalance, falls, oscillopsia, nausea, anxiety, and fear of falling.[147] As in all other sensory impairments, early intervention is recommended. Ideally, an interdisciplinary team approach should be taken and should include team members such as physicians (otolaryngologist, physiatrist, neurologist) and associated medical personnel (audiologist, physical therapist, occupational therapist, psychologist). An individualized, specific, and targeted vestibular exercise program is strongly suggested over a generic one.[16] Exercises for decreasing symptoms of dizziness or disequilibrium can be categorized into areas targeting habituation, VOR adaptation, or sensory substitution. For a patient with BPPV, canalith repositioning treatment (CRT) is often applied (Video 4.2).

Habituation

Habituation is the process by which symptoms are decreased through repeated exposure to provocative stimuli. A peripheral vestibular lesion can create a sensory discrepancy caused by unequal vestibular inputs from the vestibular labyrinths. Repetitive movement may reduce this asymmetry of vestibular input. Cawthorne[24] and Cooksey[31] treated individuals with postconcussive unilateral vestibular dysfunction and observed that patients involved in more movement gained greater management of their symptoms. They suggested that patients become more tolerant of problematic motion when exposed to repetitive exercises that progress in difficulty. Particular movements or environments that result in symptoms of moderate degree are established. Then a progressive repetitive program can be executed to help the patient habituate to the sensation, which consequently lessens the symptoms.[65]

Vestibuloocular Adaptation

The VOR is an important three-neuron vestibular reflex affected by stimulation of the SCCs that creates conjugate eye movement of equal and opposite direction to the movement of the head. The VOR allows for stabilization of the image on the retina, leading to clear vision when the head is moved. The gain of the VOR is the ratio of eye velocity to head velocity. The ideal gain in a normal patient is 1:1. A vestibular disorder can result in decreased gain in the VOR, which in turn would result in a blurred image when the head is moved rapidly. Allum et al.[5] found that patients with acute unilateral peripheral vestibular hypofunction had reductions in vertical VOR gain by as much as 66% with an average horizontal VOR gain reduction of 50% toward the involved side and 25% toward the uninvolved side of the vestibular dysfunction.

Patients with an uncompensated unilateral vestibular weakness may be prescribed exercises that focus on adaptation of the VOR. These activities consist of performing head movements while keeping a visual target in focus (e.g., a patient is asked to keep a visual target in focus while swaying the head from side to side). The exercises progress in difficulty and should be executed at rates of head turn just slower than when the visual image becomes unfocused.[65] Exercises should be performed in diverse environmental contexts, in various positions, and at different rates of head turn.

Dai et al.[32] demonstrated success in readaptation of the VOR in patients afflicted with the central vestibular disorder of mal de débarquement syndrome (MdDS). MdDS is characterized by the prolonged sensation of swaying, rocking, and/or bobbing often triggered by sea travel. The VOR adaptation exercises utilized by the researchers using full-field visual stimuli led to substantial improvement in symptoms in 70% of their patients.

Sensory Substitution

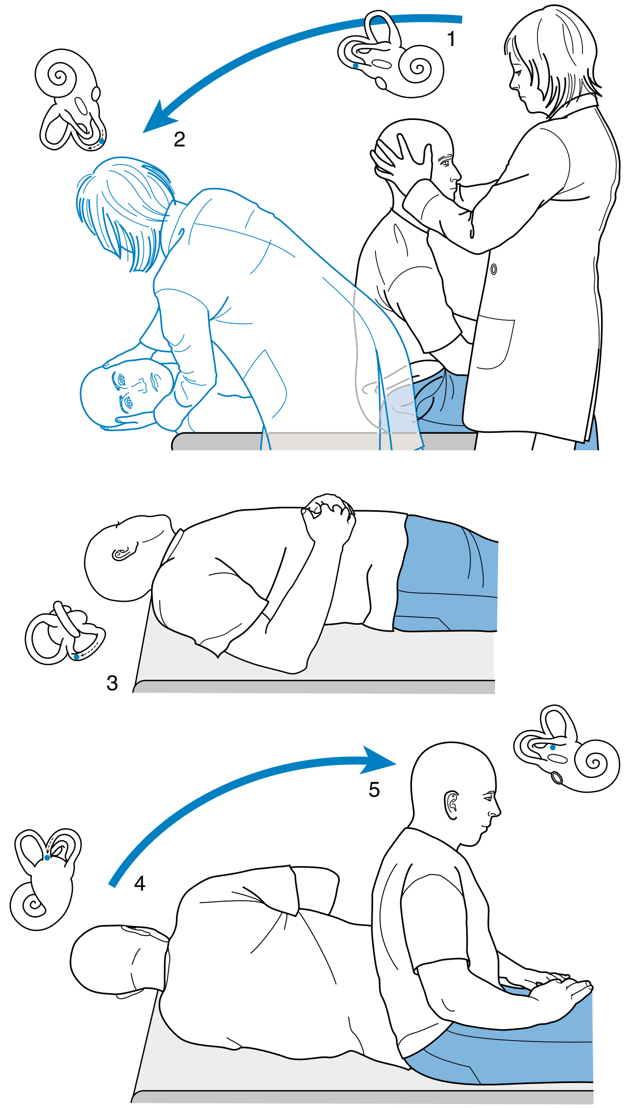
This technique involves the use of an alternative intact sense for an impaired sense. Visual and somatosensory cues are critical components of this facet of vestibular rehabilitation therapy. Enhancing the use of visual and somatosensory cues for balance and postural control may not adequately compensate for the vestibular impairment but may assist in functional recovery.

Canalith Repositioning Treatment for Benign Paroxysmal Positional Vertigo

Of the various vestibular rehabilitation interventions, the most notable in effectiveness is CRT for BPPV. This treatment, based on the maneuver described by Epley,[40] is a widely used example of CRT that is intended to move displaced particles out of an afflicted posterior SCC. The maneuver involves sequential head/body positions, with each position maintained for at least 30 seconds or until no nystagmus is observed. First, the patient sits on the examination bed with the head turned to the lesion side by 45 degrees. The patient is then lowered backward quickly until the head hangs over the plane of the bed by 30 degrees. Next, the head is rotated 45 degrees to the opposite side. Then the body is further turned away from the afflicted side until the head is positioned 45 degrees down from horizontal. Finally, the patient is returned to the sitting position. Fig. 4.15 illustrates the stages of the right-side CRT maneuver. It is well documented that BPPV can be treated successfully with the correct application of CRT.[123,130]

Efficacy of Vestibular Rehabilitation

There is sparse evidence supporting the benefits of vestibular rehabilitation in patients with most central vestibular disorders.[46]



• **Fig. 4.15** Canalith repositioning maneuver for treatment of right-sided benign paroxysmal positional vertigo. (Redrawn from Fife TDI, Lempert T, Furman JM, et al. Practice parameter: therapies for benign paroxysmal positional vertigo [an evidence-based review]: report of the Quality Standards Subcommittee of the American Academy of Neurology. *Neurology.* 2008;70:2067-2074.)

Patients with central vestibular disorders are noted to progress at a slower rate than those with peripheral disorders and thus require more time for optimal rehabilitation results.[83] Conversely, there is abundant evidence for the efficacy of vestibular rehabilitation for patients with peripheral vestibular disorders.

A systematic review on vestibular rehabilitation in populations with unilateral and peripheral vestibular disorders[67] found that vestibular rehabilitation was more effective than alternative forms of management (e.g., medication) or other control or placebo interventions in improving subjective reports of dizziness. Vestibular rehabilitation was also found to produce improvements in walking, balance, vision, and activities of daily living. Moreover, the beneficial effects were maintained in the studies with follow-up assessment. Combining the repositioning maneuvers with general vestibular rehabilitation was effective in improving long-term functional recovery. There is strong evidence to support the use of vestibular physical therapy in alleviating vestibular symptoms such as dizziness, as well in improving gaze abnormality and postural stability.[53] Based on a 2022 clinical guideline,[53] clinicians should provide vestibular rehabilitation to adults with unilateral or bilateral vestibular hypofunction when they exhibit vestibular impairments, activity limitations, and participation restrictions that are caused by their vestibular deficits. Recent advancements in habituation training, gaze stability training, balance and gait training by physical therapists, as well as innovations in virtual reality and biofeedback have also shown a positive trend in improving patient outcomes.[147]

CASE 4.2

Vestibular Impairments

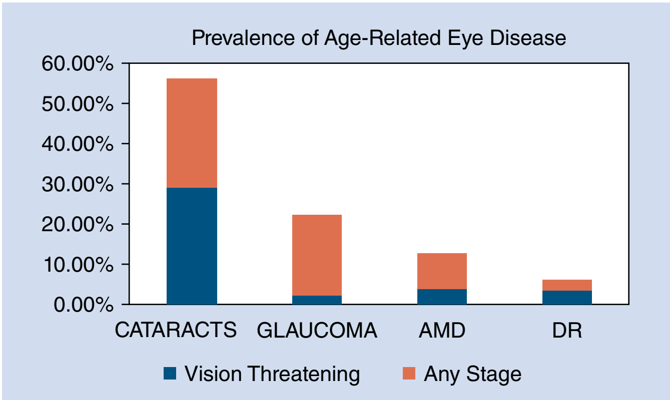
As described by Fife,[42] a 71-year-old male complained of episodic spinning, lightheadedness, and imbalance that he had experienced for several months. During the history taking, the patient reported that he had an episode of a rotational feeling when he awoke from sleep. At other times, he experienced a spinning sensation for about 20 seconds when he was reaching into an upper cabinet. He reported that his most recent episode of spinning lasted for about 10 seconds when he got out of bed and that it disappeared within 20 seconds after he sat down. His most recent spell of lightheadedness occurred while he was standing in line at a grocery store.

A cardiac workup revealed no heart abnormalities. A brain magnetic resonance imaging as well as laboratory studies (complete blood count, electrolytes, liver enzymes) were also normal. During physical examination, Dix-Hallpike positioning to the left side triggered paroxysmal positional up-beat and clockwise cyclotorsional nystagmus, indicating left-sided benign paroxysmal positional vertigo (BPPV).

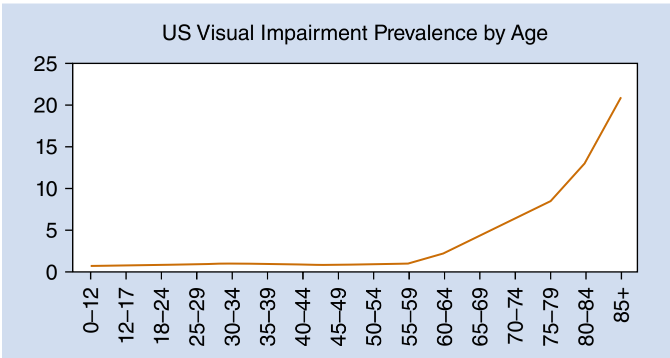
Based on the diagnosis of BPPV, canalith repositioning treatment was performed. All symptoms and nystagmus disappeared with repeated repositioning. However, at 3-week follow-up, the patient reported episodic dizziness while standing up or walking. The Dix-Hallpike maneuver was performed again and found to be normal with no vertigo or nystagmus this time. Gradual declining of blood pressure (nadir of 75 mm Hg systolic) at 20 minutes of head-up in association with lightheadedness was found during a tilt table study, indicating delayed orthostatic hypotension. Normal heart rhythm was observed. His antihypertensive medications were adjusted, and low-dose fludrocortisone was prescribed for eventual resolution of his lightheadedness.

Visual Impairments

Visual impairment (VI) can result from damage anywhere along the visual pathway from the globe back to the visual cortex and forward to the sensory association areas. Visual dysfunction is very common in the United States, with the number of visually impaired individuals estimated at over 7 million, including 1.08 million who are legally blind.[43] (The data exclude individuals who function as visually impaired because of uncorrected refractive error.) The most common cause of VI is age-related eye disease, including cataracts, glaucoma, macular degeneration, and diabetic retinopathy (DR)[70] (Fig. 4.16). However, vision loss can occur at any age because of hereditary disorders, ophthalmic injury, or brain injury. Among those under 40 years of age, there are an estimated 1.6 million cases of VI and 141,000 cases of blindness. Current estimates show 0.74% of those under 12 years old, 0.99% of those age 50 to 54, and 20.73% of those aged 85 and older are visually impaired[43] (Fig 4.17). According to the National Eye Institute, VI is classified as "any chronic visual deficit that impairs everyday functioning and is not correctable by ordinary eyeglasses or contact lenses."[102] VI is categorized by the vision in the *better seeing eye* (Table 4.2). The Social Security Administration defines legal blindness as (1) best corrected visual acuity of 20/200 or less *in the better seeing eye* or (2) a visual field limitation such that the widest diameter of the visual field in *the*



• **Fig. 4.16** Prevalence of age-related eye disease pathology differentiated by total incidence vs. vision threatening.[70] *AMD*, Age-related macular degeneration; *DR*, diabetic retinopathy.



• **Fig. 4.17** Visual impairment prevalence by age in the United States. (Redrawn from Flaxman AD, Wittenborn JS, Robalik T, et al. Prevalence of visual acuity loss or blindness in the US: a Bayesian meta-analysis. *JAMA Ophthalmol*. 2021;139[7]:717-723.)

TABLE 4.2 Categories of Visual Impairment

Impairment Category	Visual Acuity
Mild Visual Impairment	Better than 20/70 in the better seeing eye
Moderate Visual Impairment "Low Vision"	20/70 to 20/200 in the better seeing eye
Legal Blindness	20/200 or worse OR visual field 20 degrees or less in the better seeing eye

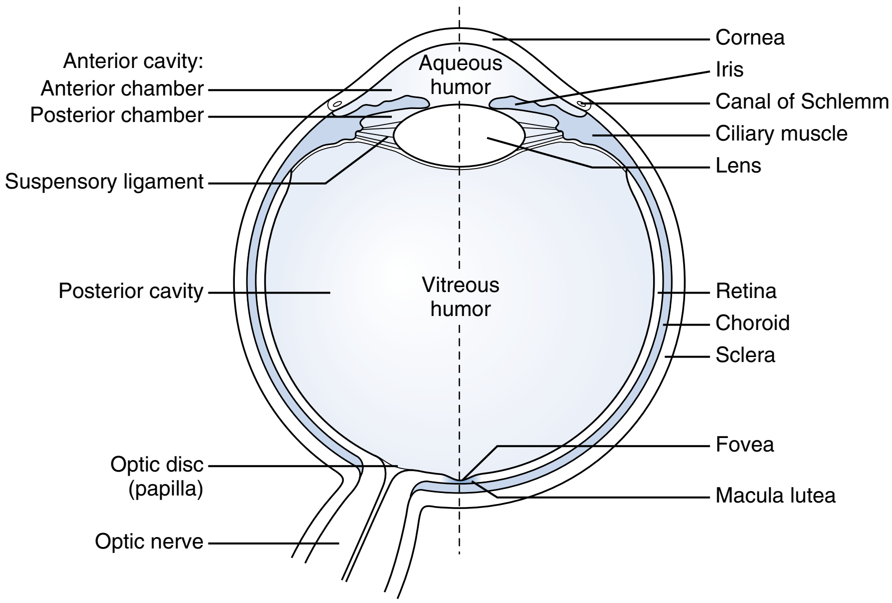
better eye subtends an angle no greater than 20 degrees. This criterion defines eligibility for most rehabilitation services offered by government and private agencies.

Anatomy of Vision

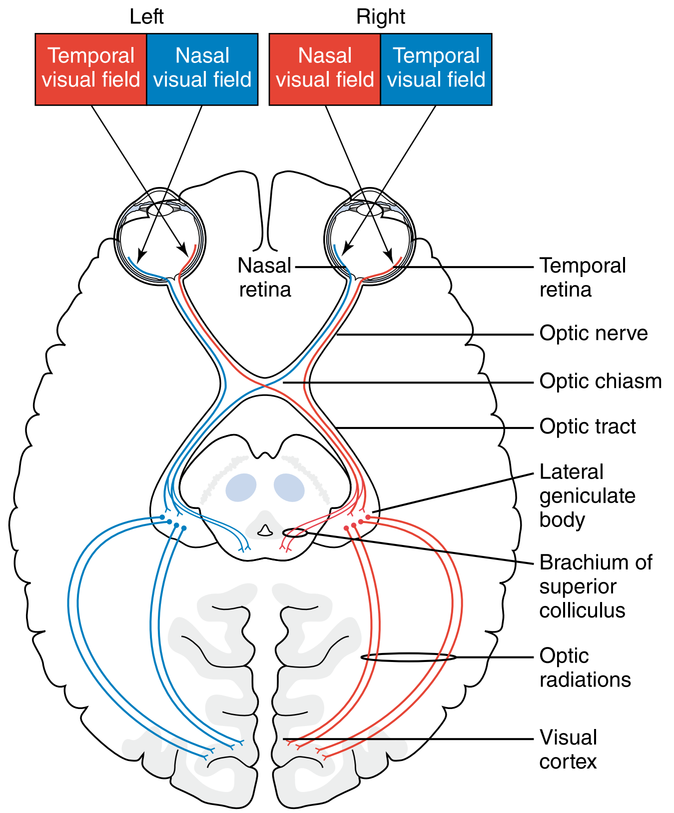
Fig. 4.18 shows the structures of the eyeball. Light enters the eye through the cornea and travels through the lens. Both these structures function to bend light into focus on the retina. Light that comes into focus before the retina is termed myopia, and light that would come into focus beyond the retina is termed hyperopia. Presbyopia is the age-related loss of accommodative ability to see at near distances. Light lands on the outer layer of the retina, which contains rod and cone photoreceptor cells. At the optical axis of the eye, the retina has a specialized region in the macula known as the fovea, which contains densely packed cone photoreceptors. The fovea, although small in area (comprising only about 1% of the retina), is disproportionately represented in the visual cortex (cortical magnification), allowing for resolution of fine detail. Rod and cone photoreceptors synapse onto horizontal and bipolar cells. The bipolar cells synapse onto amacrine and retinal ganglion cells. The retinal ganglion cells come together to form the optic nerve exiting the eye at a single point and travel back to the lateral geniculate nucleus (LGN). Along the pathway from the globe to the LGN, the optic nerves meet at the optic chiasm. At this location, the nasal nerve fibers from each eye will cross, and the temporal fibers will remain ipsilateral. This organization will allow all images seen in the left visual field of both the left and the right eye to come together in one bundle known as the right optic tract and vice versa for the right visual field going to the left optic tract. This organization results in the left half of each eye's visual field being transmitted in the right hemisphere of the brain for the entire visual pathway beyond the chiasm. At the LGN, the optic tract synapses with the optic radiations, which have separate pathways that either go up through the parietal lobe or down through the temporal lobe on the way back to the occipital lobe. The radiations terminate at the visual cortex, V1, of the occipital lobe. Fig. 4.19 shows the visual pathway from the retina to V1. V1 is the primary visual cortex, also called the striate cortex. It receives visual information and responds to orientation and direction. This information is then sent forward to the prestriate cortex, extrastriate cortex, and associative areas that help the brain respond quickly to patterns, spatial information, and learned visual information without additional effort. After leaving V1, visual processing travels through the ventral and dorsal streams, commonly known as the "what" and the "where" pathways. The ventral stream goes in the direction of the inferior/temporal cortex, and sites along this pathway are associated with color, facial, form, and object recognition. The dorsal stream goes in the direction of the middle temporal area and posterior parietal cortex detecting spatial information, location, and motion and steering visually guided body movements. Table 4.3 reviews higher-level visual processing disorders that can occur when there is damage to the ventral or dorsal streams.

Risk Factors for Developing Vision Impairments

Risk factors for developing VI include uncorrected refractive error, age-related eye disease, hereditary eye disorders, globe/orbit



• **Fig. 4.18** Structures of the eyeball. (Redrawn from Bhatnagar SC. *Neuroscience for the Study of Communicative Disorders.* 4th ed. Lippincott Williams & Wilkins; 2013.)



• **Fig. 4.19** The visual pathway from the retina to the visual cortex. (Modified from Bhatnagar SC. *Neuroscience for the Study of Communicative Disorders.* 4th ed. Lippincott Williams & Wilkins; 2013.)

trauma, and brain injury. The most common etiologies are discussed next.

Uncorrected Refractive Error

Refractive error is a correctable form of VI. Despite available interventions, it is still a major cause of VI in the United States, with 73% to 83% experiencing functional VI caused by uncorrected refractive error.[25,158]

Cataracts

A cataract is a gradual clouding of the crystalline lens within the eye, which leads to a progressive decrease in vision. Risk factors for cataract development are age, smoking, diabetes, hypertension, obesity, corticosteroids, history of ocular trauma or surgery, and ultraviolet light exposure.[3,23,35,77,107,160,169] In the United States, cataracts are a reversible form of blindness because of the high availability of cataract extraction and government insurance programs that largely cover the cost of cataract surgery. Barriers remain patient awareness, choice, and financial.

Glaucoma

Glaucoma is a group of eye diseases that cause damage to the optic nerve. As retinal ganglion cell axons that make up the optic nerve are lost, visual signals traveling from the retina to the brain decrease, creating visual field loss. Glaucomatous visual field loss occurs in typical patterns; most commonly it starts peripherally, specifically nasally, and moves in an arcuate pattern temporally, sparing central vision. In some types of glaucoma, it may start paracentral. Uncontrolled glaucoma can progress centrally, creating a decreasing tunnel of vision and may result in total blindness. Risk factors include age, family history, Black or Hispanic race, type 2 diabetes, and ocular features of high intraocular pressure, thinner corneas, lower ocular perfusion pressure, and history of ocular injury.[99,124] Treatment may include pharmacologic, laser, or surgery. Goals of treatment are to halt the glaucomatous progression; nerve loss that has occurred cannot be regenerated.

Age-Related Macular Degeneration

In general, there are two types of age-related macular degeneration (AMD): dry/nonneovascular and wet/neovascular. About 85% to 90% is dry AMD characterized by drusen (accumulations of extracellular material) and abnormalities in the retinal pigment epithelium.[45,74] At this stage, patients generally have milder symptoms, but about 12% of these patients will suffer severe vision loss due advanced geographic atrophy where extensive photoreceptor cells are lost.[45] Wet AMD accounts for 10% to 15% and is characterized by the development of neovascularization leaking blood and/or plasma.[45,74] Patients generally experience more severe vision loss at this stage. AMD has a predilection for the macula and therefore creates a loss of central vision leaving peripheral vision intact. Patients can experience metamorphopsia, central scotomas, and rapidly deteriorating vision. Risk factors for AMD are aging, White race, smoking, hypertension, hypercholesterolemia, cardiovascular disease, female sex, and genetics.[45,74] Treatment may include high-dose vitamins and antioxidants, dietary and environmental recommendations, and antivascular endothelial growth factor (anti-VEGF) intravitreal injections.[38,114,146,166] The development of anti-VEGF intravitreal injections has been transformative in the management of wet AMD, allowing patients to maintain clearer acuity.[146,166]

Diabetic Retinopathy

DR occurs when there is a breakdown of the blood-retinal barrier in the retinal capillary basement membranes. DR is divided into nonproliferative DR (NPDR) and proliferative DR (PDR) based on the presence or absence of neovascularization. The majority of DR is NPDR with PDR occurring in about 5% to 10% of cases.[74,116] Patients with DR generally have no symptoms until late stages, at which time the vision loss can be rapid and severe. In NPDR, nearly all cases are asymptomatic unless macular edema evolves, at which time central vision blurs. In PDR, neovascularization can hemorrhage into the macular preretinal space or the vitreous, which blocks the path of light, causing severe loss of vision. This type of vision loss may spontaneously resolve in weeks to months, but severe and permanent loss of vision can result from macular ischemia, tractional retinal detachments, or neovascular glaucoma. Risk for DR is duration of diabetes, aging, glycemic control, hypertension, hyperlipidemia, diabetic nephropathy, and pregnancy.[27,41,74,82,170] It is important to screen diabetic patients with a dilated eye exam regularly because early detection and treatment can lessen the risk of severe vision loss. Treatment may include diet/lifestyle recommendations, pharmacological use, anti-VEGF injections, laser, and surgery.

Acquired Brain Injury

Two of the most common types of acquired brain injuries are cerebrovascular accident (CVA) and TBI. According to the Centers for Disease Control and Prevention, more than 795,000 people in the United States suffer a CVA each year, and 2.8 million have a

TABLE 4.3 Visual Perception Disorders

Disorder	Definition	Site of Typical Causative Lesions
Central Hemiachromatopsia	Inability to perceive color in one hemifield	*Ventral Stream* V4: inferior occipital cortex
Visual Agnosia	Inability to recognize objects: – Apperceptive: cannot perceive the geometric shape, or orientation of an object, cannot copy or recognize it – Associative: can perceive and copy object shape but cannot recognize it	*Ventral Stream* – Apperceptive: left occipitotemporal region – Associative: left temporal
Pure Alexia (Alexia without Agraphia)	Inability to recognize or read text, but maintains ability to write	*Ventral Stream* Left occipital cortex and crossing fibers in splenium of the corpus callosum OR visual word form area in the fusiform gyrus
Prosopagnosia	Inability to recognize faces	*Ventral Stream* Fusiform gyrus of temporal lobe
Topographagnosia (Landmark Agnosia)	Inability to recognize familiar landmarks/scenes	*Ventral Stream* Parahippocampal place area
Akinetopsia	Impaired ability to detect visual motion; "motion blindness"	*Dorsal Stream* Parietooccipital region, MT/V5
Visual Neglect	Perceptual loss of contralesional hemisphere	*Dorsal Stream (especially the right hemisphere)* Parietal lobe: visuospatial Temporal lobe: allocentric Frontal lobe: visuomotor
Simultagnosia	Inability to perceive the global precept of a visual scene despite being able to see individual elements	*Dorsal Stream* Parietal lobe *Balint syndrome: bilateral parietooccipital lesions causing simultagnosia, optic ataxia, and ocular apraxia
Optic Ataxia	Impaired reaching for targets using visual guidance	*Dorsal Stream* Parietal to premotor cortex
Ocular Apraxia	Inaccurate saccades	*Dorsal Stream* Connections between parietal, superior colliculus, and frontal eye fields

From Haque S, Vaphiades MS, Lueck CJ. The visual agnosias and related disorders. *J Neuroophthalmol.* 2018;38(3):379-392; Prasad S, Dinkin M. Higher cortical visual disorders. *Continuum (Minneap Minn).* 2019;25(5):1329-1361.

TBI.[150,154] The extensive neural network of the brain, serving the visual system and its interconnections with other sensory systems and higher order functions, leaves the visual system vulnerable to axonal injury, atrophy, and neuronal circuit disruption. Common VIs after these injuries include hemianopsia (CVA 8%–57%/TBI 18%[48,97,119,120,125,132,134,135]), visual neglect (CVA 14%–23%/TBI 5%–9%[30,50,117,132,135]), cranial nerve palsy (CVA 10%–37%/ TBI 1%–7%[13,29,135]), pursuit/saccade versional deficits (CVA 5%–38%/TBI 7%–39%[6,29,30,47,49,132,135,144]), convergence insufficiency (CVA 33%–55%/TBI 23%–50%[6,47,59,94,98,153,168]), accommodative dysfunction (CVA 13%/TBI 31%–47%[29,30,47,49,97]), and photosensitivity (CVA 5%–30%/TBI 7%–55%[6,49,59,94,98,153,168]). Table 4.4 provides descriptions and symptoms of common VIs found in acquired brain injury. A thorough visual examination is recommended after an acquired brain injury because vision disturbances occur at all severities of brain injury.

Vision Assessment

The visual assessment for the patient with low vision and brain injury can begin with the rehabilitation provider. Box 4.6 provides case history questions, and Table 4.5 lists visual assessment screening techniques.

Eye examinations should regularly be scheduled for any patient with low vision or acquired brain injury. Eye exams in these patients should consist of two components: (1) the dilated eye health exam and (2) the vision rehabilitation (VR) exam, which may be geared toward low vision or binocular vision. The low vision examination is provided by a low vision optometrist or ophthalmologist. A binocular vision exam may be provided by neuro-optometry, behavioral optometry, neuro-ophthalmology, strabismus specialist, or (if lack of resource exists) a low vision clinician. VR providers frequently work alongside blind rehabilitation specialists (certified: vision rehabilitation therapists, low vision therapists, orientation and mobility specialists), occupational therapists (with specialty certification in low vision), and/or vision therapists.

Vision Rehabilitation

VR focuses on restoring maximal function for a visually impaired individual, including increasing independence, sense of well-being, and quality of life using restitution, compensatory, and

TABLE 4.4 Common Visual Impairments Found in Acquired Brain Injury

Impairment	Description	Symptoms
Homonymous Visual Field Defect	Hemianopsia or quadranopsia, which may be congruent or incongruent, partial or complete	Difficulty with navigation, bumping into objects or people on one side Difficulty with reading, computer, television
Visual Neglect	Perceptual loss of contralesional visual field	Patients do not report difficulty. Their family or care team may notice a lack of awareness or engagement with their contralesional side.
Cranial Nerve 3 Palsy Cranial Nerve 4 Palsy Cranial Nerve 6 Palsy	– Eye rests down and out Ptosis and paralysis of adduction, elevation, and depression. Pupil may be dilated. – Eye rests elevated. Paralysis of depression in adducted gaze. Contralateral head tilt. – Eye rests in. Paralysis of abduction.	Diplopia: constant or in specific gaze Slanted vision Dizziness Blurred vision
Convergence Insufficiency	Difficulty turning the eyes inwards to view an object at near	Asthenopia and headaches with reading Intermittent blur/intermittent diplopia Ill-defined dizziness
Accommodative Insufficiency	Difficulty focusing at near	Asthenopia and headaches with reading Blurred vision at near
Versional Deficit: Pursuit or Saccade Dysfunction	Pursuit: Smooth eye movement that tracks a slow-moving target. Saccade: Rapid eye movement to reorient gaze toward an area of interest	Frequently loses place reading Skips words or lines Excessive head movements Slow reading/poor comprehension

• BOX 4.6 Case History Questions

Brain Injury

- Any changes with your vision after stroke/traumatic brain injury?
- Have you lost any of your side/peripheral vision?
- Do you experience any double or blurry vision?
- Are you having difficulty reading?
- Do you have difficulty with glare?

Low Vision

- Any difficulties with near tasks (reading, writing, medication labels, telling time)?
- Any difficulty with distance tasks (watching television, reading street signs)?
- Do you have difficulty with glare?
- Is strong lighting essential for reading?
- Any falls/trips or unwanted contacts because of something you did not see?

substitution strategies.[20] Restitutive treatment aims to bring visual function to its highest remaining potential, ideally back to normal. In the setting of brain injury, time is restitutive, aligning with the restoration of normal neuronal physiology.[119] Compensatory treatment uses adaptive strategies to minimize the impact of the VI. Substitution treatment uses adaptive aids to function with the VI. Rehabilitation is tailored to an individual's goals. Strategies will focus on whether the deficit involves central, peripheral, or cortical vision loss.

Central vision loss, which occurs in pathologies such as macular degeneration or diabetic macular edema, initially impact near vision tasks, most commonly reading. Later, distance vision symptoms become more noticeable with difficulty driving, watching television, and glare.

The most frequently used rehabilitation strategies utilize optical devices (magnifiers, telescopes, microscopes), nonoptical devices (task lighting, talking watch, tinted glasses, large print items [calendars], meal prep supplies, bold pens, bold-lined paper), and electronics (closed-circuit televisions, computers, smartphones). Smartphones, tablets, and computers have software available for all levels of VI, including magnifiers, text-to-speech, and voice commands. Patients with central vision loss may benefit from eccentric view training, which is a compensation strategy utilizing an area of the retina away from the damaged fovea to improve visual functioning.[69] Rehabilitation has been shown to be effective in restoring function and independence.[96,111]

Peripheral vision loss, which occurs in pathologies such as glaucoma, hereditary disorders such as retinitis pigmentosa, or homonymous hemianopsia, initially tend to affect mobility, object detection, driving, and glare sensitivity. If the condition evolves to include the central field, the central visual loss symptoms described earlier become apparent. The most frequently used rehabilitation techniques for peripheral field loss are orientation and mobility aids (long-white guidance cane), lighting, peripheral prisms, and tinted glasses. Home safety evaluations with recommendations for minimizing tripping hazards are also beneficial. For homonymous hemianopsia, especially right sided, or if peripheral visual field loss has constricted to the extent that 15 to 25 characters cannot be viewed simultaneously (inhibiting effective reading saccades and text navigation), text-to-speech technology can be a very helpful adjunct reading strategy.[91,92,101,128,129,172] The National Library Service offers a free talking book library to those who are visually impaired; registration materials can be found at https://www.loc.gov/nls/. Peripheral sectoral prisms placed above and below the line of sight can achieve a 30-degree expansion into the blind field with a moderate acceptance rate.[18] Patients with peripheral visual field loss may also benefit from

TABLE 4.5 **Visual Assessment Screening**

Test	How Performed	Expected Outcome
Visual Acuities Distance and Near	Patient should wear habitual distance and near glasses. If unavailable, a pinhole occluder can be used to estimate best corrected acuity at distance.	20/20 at distance and near If unexplained acuity loss, refer for eye exam. If 20/50 or worse best corrected, refer to low vision.[a]
Pupils	Observe pupils for direct and consensual responses in dim and bright lighting. Perform swinging flashlight test for relative afferent pupillary defect (RAPD).	Equal, round, reactive, -APD If new RAPD or asymmetric pupils, refer for eye exam.
Extraocular Muscles (EOMs)	Have patient look in the six cardinal gazes. If the EOM appears restricted in any gaze, perform monocular *ductions* rather than binocular *versions.*	Full and equal amplitude of motion between the two eyes If restriction is noted or patient is reporting diplopia, refer to vision rehab.
Confrontation Fields	Check ability to count fingers in all four quadrants of each eye and check for simultaneous perception. Perimetry should be performed on all patients with neurologic lesions that could produce a field cut.	Full in both eyes +Simultaneous perception If visual field loss is suspected, refer to vision rehab.[b]
Near Point of Convergence Assessment	Patient should use near glasses. The patient is asked to report when they see double as a 20/30 size target is moved toward the bridge of the nose at eye level. Break point is noted when the patient reports diplopia or when the examiner notes that one eye has stopped tracking the target and has drifted out. The target is then moved away until the patient reports single or the examiner notes realignment.	5 cm break/7 cm recovery Refer to vision rehab if patient has a receded near point of convergence and has symptoms of convergence insufficiency.
Accommodative Amplitude Pull away method Assessment	Patient should use distance glasses. Test monocularly. Start by holding a 20/30 size target near the eye and slowly pull away asking the patient to report when the letter has become clear. Record in metric system. 1/meter = diopters	Minimum expected amplitude in diopters = 15 − (¼ × age) Refer to vision rehab if accommodation is worse than age-expected norms or if patient has symptoms of accommodative insufficiency.
Pursuit Eye Movement Assessment	Hold target 40 cm away from the patient and rotate target in two slow clockwise, followed by two slow counterclockwise circles of 20 cm diameter. Observe for ability, accuracy, and head/body movement.	Completes all rotations with two or fewer refixations and no more than slight head/body movement. Refer to vision rehab if unable to track slow-moving target or if symptomatic.
Saccadic Eye Movement Assessment	Hold two targets 40 cm away from the patient and spaced 20 cm apart. Have the patient alternately look from one target to the other for five round trips. Observe for ability, accuracy, and head/body movement.	Completes five round trips with no more than slight undershooting or head/body movement. Refer to vision rehab with poor saccadic accuracy or if symptomatic.

[a]Most states require 20/40 vision for an unrestricted driver's license.

[b]Visual field requirements for a driver's license vary widely by state from "no" visual field requirement to 140 degrees. Most states require 110 to 140 degrees, making patients with complete homonymous hemianopsias ineligible for a driver's license.

From Scheiman M, Wick B. *Clinical Management of Binocular Vision: Heterophoric, Accommodative, and Eye Movement Disorders.* 3rd ed. Lippincott Williams & Wilkins; 2008.

scanning training, a compensatory strategy utilized to improve ability to use remaining vision more effectively by increasing the number and amplitude of saccades into the blind field in an organized pattern. Such techniques can help improve mobility, avoid obstacles, and increase reading speed.[17,51,60,95,112,115,172]

Acquired brain injury–related vision loss, which includes strokes and TBI, tends to affect visual fields, binocular vision, and visual perception. Symptoms are described in Table 4.4. While many rehab intervention studies show preliminarily positive results, effectiveness is not firmly concluded because of low-quality studies. Interventions can be focused on restitution, compensation, or substitution strategies with techniques frequently combined.[57]

Eye movement impairments include vergence, accommodation, and versions. Oculomotor rehabilitation has demonstrated preliminary good results for vergence and accommodative dysfunction with the most consensus for orthoptic vergence exercises.[2,47,66,100,127,136,139,140,142,151,152] There is insufficient evidence for versional oculomotility training; however, improved accuracy and reading ability have been reported in limited studies.[52,75,133] Additionally, while not restorative, accommodative insufficiency can be addressed with addition lenses or large print. Convergence insufficiency can potentially be managed with prism lenses or monocular occlusion at near.

Visual field defects/neglect have a growing body of evidence that suggests scanning training can help an individual develop more effective search patterns. Exercises that focus on increasing saccadic amplitude, velocity, and fixation as far into the blinded hemifield as possible is one effective strategy.[39,76,95,109,113,115,131,164] A Cochrane review found stronger evidence for improving visual exploration than for reading with scan training.[120] Several studies and systematic reviews have suggested that rehabilitation with prism adaptation, visual scanning, and transcranial magnetic stimulation may be effective in visual neglect, but further randomized controlled trials eliminating risk of bias are needed.[7,9,11,28,93,159,163,165] As stated in the peripheral vision loss section earlier, peripheral prisms, text-to-speech low vision technology, and orientation and mobility can also be utilized.

For eye alignment, such as for ocular cranial nerve palsies, use of prism lenses, occlusion, or strabismus surgery can be considered.[79,80] Prism is an accepted and effective treatment intervention for diplopia. If the eye deviation is too large for prism, occlusion with eye patch, Bangerter foil, or frosted tape can be utilized. Sectoral occlusion, to eliminate diplopia in a particular direction of gaze while maintaining binocularity elsewhere, can be considered. In the setting of cranial nerve palsy secondary to acquired brain injury, strabismus surgery should be postponed until recovery has ceased and eye misalignment is stable. Patients can also be taught compensatory head postures to avoid utilizing direction of gaze that creates diplopia.

Dual Sensory Impairment

Activity limitations through normal aging processes (e.g., diminished hearing, increased fall risk) may magnify the effects of sensory loss, including vision and hearing loss. A systematic review of the literature suggests that the prevalence of dual sensory loss (hearing and vision) among individuals 60 years and older varies widely, from 3.3% to 64%, depending on definition, measurement, sampling, and methodology.[61] Among all age groups, dual sensory impairment (DSI) it is most prevalent in persons 80 years of age and older[21,148] but is also a growing concern for the younger post-9/11 veteran population. In an outpatient sample of veterans aged 44 to 95 years, who received both auditory and visual clinical evaluations in a Department of Veterans Affairs (VA) hospital, DSI ranged from 5% to 7.4%.[145] Blast injuries and TBI can have widespread effects to multiple body systems that impact the ability for the patient to fully participate in recovery.[138] Among a sample of outpatient post-9/11 veterans with blast exposure and TBI history, self-reported DSI was documented in 34.6% of the cohort.[88] DSI has been shown to impede rehabilitation among veterans receiving inpatient treatment at a VA polytrauma rehabilitation center as well as among civilians.[19,87]

Psychological Adjustment

Vision loss has a holistic effect, including decreased quality of life, depression, familial stress, social isolation, and economic stress, which, in combination, may ultimately affect mortality risk.[39,78,171] Therefore the individual's adjustment to vision loss should be addressed. Referral to local psychosocial support services should be strongly considered. Support groups have also been shown to improve quality of life, with those individuals with initial lower quality of life scores showing the greatest improvement.[84] Improved adjustment to vision loss entails acceptance of the condition, a positive attitude toward adjustment, and social support from family, friends, and peers in addition to participation in rehabilitation services.

Summary

This chapter provides descriptions of the diagnosis and rehabilitation of three of the most common sensory impairments—those in the auditory, vestibular, and visual domains. For the human brain to properly process incoming signals for communication and balance, these three systems must work in synchrony.[44] DSIs (affecting both the auditory and visual domains) can have devastating effects on the person's ability to compensate and to participate in the rehabilitation process.[87–89] Multisensory impairments (involving auditory, vestibular, and visual domains) may also compromise the recovery process and should be addressed early by the rehabilitation team.[86,118] Clinicians and researchers must work together to provide early diagnoses and rehabilitation of these common sensory impairments so that the persons served can eventually achieve maximal function and quality of life.

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